# Outline (until the next coffee break)

- Lecture: Modeling of regulatory activity with sequence models
- Hands-on: Training a sequence-based model
- Discussion of pre-training and fine-tuning approaches
- Hands-on: Interpreting models with in-silico mutagenesis

IGVF

# Sequence modeling

**ISMB-ECCB 2025:** *Tutorial IP2 (MPRAs and IGVF Resource)*

**Arjun Devadas**
**Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH**

July 20, 2025

UNIVERSITÄT ZU LÜBECK
UKSH UNIVERSITÄTSKLINIKUM Schleswig-Holstein
BIH Berlin Institute of Health @Charité

# Scope of this session

## Goals

- Intuitive understanding of machine learning and basic sequence models
- Neural networks (NNs): What are they? and How they work?
- Example: **DeepSTARR**
- Model Interpretation

## Questions we will cover:

- Why sequence-based modeling?
- How to leverage NNs alongside experimental data?
- What can we gain from NNs and how?

# Machine learning 101

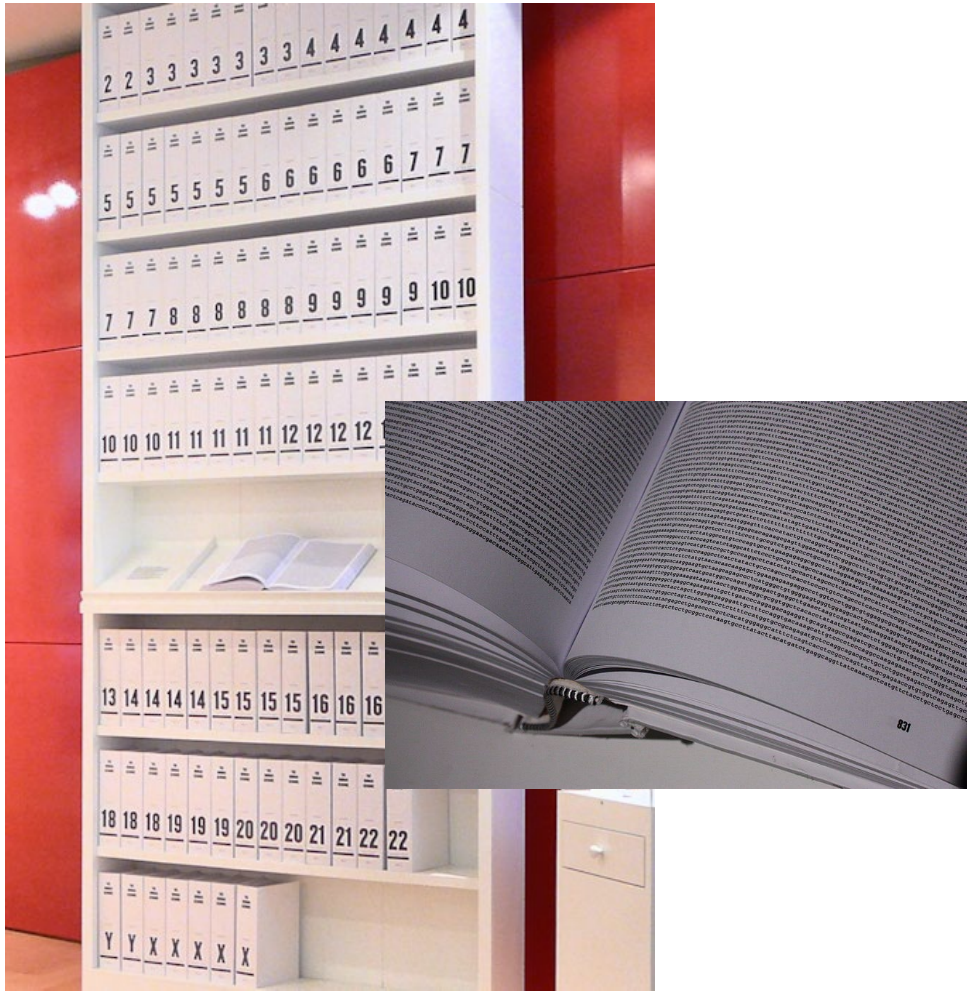

https://xkcd.com/1838/

make computer systems … learn generalizations … (without explicit information) … from patterns of data

# Basic terminology

- **Data – typically transformed to derive features**
- **Typically, a data set (<u>features</u> + output/<u>response</u>) is split into:**
  - **train and test set**, entries within these sets are called **examples**
  - sometimes an (independent) **validation set** is used, and the **test set serves for optimizing hyperparameters** (i.e., all kinds of tuning parameters to the model, like the number of parameters or training iterations)
- **Responses can be categorical (<u>classification</u>) or numerical (<u>regression</u>)**
- **Model training is either:**
  - <u>Supervised:</u> response is available, and the model is trained to produce the response
  - <u>Semisupervised:</u> proportion of training examples does not have a response
  - <u>Unsupervised:</u> clustering (e.g. kmeans, graphs/cliques, self-organizing maps)
- **Model performance is assessed by comparing predicted responses with original responses**
- **Models can be trained on surrogate data / correlated problems, recently called “zero-shot”**

# Sequence modeling



Printed Human Genome of the Wellcome Collection

```
CCTGGTGGAAGAATTTAGTGACATTTGTTGGCTGCTTCTTAGGGAAGAAACCAAAGTCTGAC
TCTAACTCCACAGTATTTTCCCCCTGGTGACTGACACAGCTGACCTAGAGCTAGAAGTCACC
TCATCTGACCCCTGGTTGCTGGCTCTCCACTTGACAATGGAGAGGACATGGACCCTGAGGGC
CTCTCCCACACAGACTGGGCCTCACCTCTCCACAGAGGAATTCAGCTCAGGAGTAGATGTAC
AGATGCCGGCACGATCTGCTTACAGCTTCTCATTGCAGGCATAGTCTTTTTTTTTTTTTTT
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AGCACTTGAGTAGTAATTGGGGTGTGGTGAGGGAGAAAACAAATGCCTCAAGGGGGCAGCCA
CAGCCCCTAACTCTAGGCCTGACAGAGAAGGCAGGGAGCTCGGCAGACAGGTGCGTGGAGAC
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CTGTCCTCAGCACAGAGGCAAGGTCCTCAATAAAGAGCTGGGAGCAGTGACTGCAGACTCAG
AGGGTCTCTTCTCCTGGATCCTTATCTGGCTTCCACCCCAGGAAGCCAGGCCATTGCTACAC
CTGTGTGAGGCTCTCTAGGGAGATGGGGCTGGGGCTGGCCCTGCTTCCCTTCTGCATTTATC
AAGTGCCTGTGACAGGGAAGACACCACAAGAGGGTCACCGACCTGAGGGCAGTGCAGTCCCA
GGTCTTGAAATAGTGAAGTTCAGTTTCTAAACCCCTCTGTGCCTCCTTCTTGCTCTCTGTAA
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GTTTCCAAGCTTTCTGAAAAATGAGGCCCCTTATCAATGTTGAAAAATAGGTAAATCTCATT
CTTTTTTATGACTGAACTTAAAGAATGTGATTGGATTGTTTGTAACCCAAAGTATAAAGGCT
TTTTTTTTTTTTTCTGAGACAAGGTCTCACTCTGTCACCTAGGCTGGATTGCAGTGGCATAA
TCACAGCTCACTGCAGCCTTGACCTCCCAGGCTCAGGCGACCTTCCTGCCTCAGCCTCCTGA
ATAGCTGGGATTATAGGTGTGCATCACTATGCCTAGTTAATTTTTGTATTTTTTGTAGAGAC
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GGGTGGTGGCAACACTATGAGCTGTTGGGAAAATATAACTAAATCTATTATTAAGTTTAAGT
AAAATAATTGAATATTATATAATTATATGAATTAATCAAATATAATTAAATATATTTCATAA
ATAAATATTATTAGAAATAATTCAAGGAGAAAAAAAATCCCCCAACCCAGAAAAACCTGAGG
```
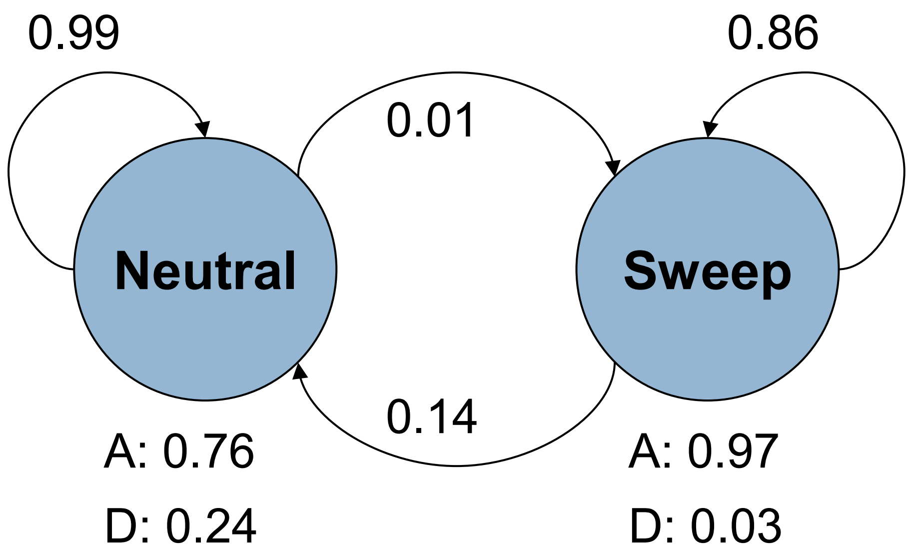

# A classic – Hidden Markov Models (HMMs)

- **E.g. two state model of genomic regions under neutral evolution vs selection, based on sequence of single nucleotide polymorphisms in an ancestral (A) or derived (D) state (similar to "Occasionally dishonest casino" with fair vs loaded coin)**
- **HMMs can be used to:**
  - Predict gene models, e.g. define Exon/Intron boundaries
  - Segment the genome based on sequence and/or sequence annotations (e.g. ChromHMM, Segway)



# kmer representation

```
CCTGGTGGAAGAATTTAGTGACATTTGTTGGCTGCTTCTTAGGGAAGAAACCAAA
GTCTGACTCTAACTCCACAGTATTTTCCCCCTGGTGACTGACACAGCTGACCTAG
AGCTAGAAGTCACCTCATCTGACCCCTGGTTGCTGGCTCTCCACTTGACAATGGA
GAGGACATGGACCCTGAGGGCCTCTCCCACACAGACTGGGCCTCACCTCTCCACA
GAGGAATTCAGCTCAGGAGTAGATGTACAGATGCCGGCACGATCTGCTTACAGCT
TCTCATTGCAGGCATAGTCTTTTTTTTTTTTTTTTTTTGGCTACTGAGCTCATCA
AAGCTGGTGAGGACCCAGCTAAGGTTGTTTGCTGCTGCTGTGAGCACTTGAGTAG
TAATTGGGGTGTGGTGAGGGAGAAAACAAATGCCTCAAGGGGGCAGCCACAGCCC
CTAACTCTAGGCCTGACAGAGAAGGCAGGGAGCTCGGCAGACAGGTGCGTGGAGA
CATTCTGATCCCAATCTCACTTAAGGATTAAGGGAGGGGGCAGTAGGGAGACAGG
ATAGCCAGTCTTGTAAATTGAGACGGGCCCAGGATCCCTTATCCTGCAATCTGTC
CTGCTGGACAGAGCACTGTCCTCAGCACAGAGGCAAGGTCCTCAATAAAGAGCTG
GGAGCAGTGACTGCAGACTCAGAGGGTCTCTTCTCCTGGATCCTTATCTGGCTTC
CACCCCAGGAAGCCAGGCCATTGCTACACCTGTGTGAGGCTCTCTAGGGAGATGG
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GACACCACAAGAGGGTCACCGACCTGAGGGCAGTGCAGTCCCAGGTCTTGAAATA
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AATCTCATTCTTTTTTATGACTGAACTTAAAGAATGTGATTGGATTGTTTGTAAC
CCAAAGTATAAAGGCTTTTTTTTTTTTTTCTGAGACAAGGTCTCACTCTGTCACC
TAGGCTGGATTGCAGTGGCATAATCACAGCTCACTGCAGCCTTGACCTCCCAGGC
TCAGGCGACCTTCCTGCCTCAGCCTCCTGAATAGCTGGGATTATAGGTGTGCATC
ACTATGCCTAGTTAATTTTTGTATTTTTTGTAGAGACGGGGCTTCATCATGTTGC
CCAAATGCTTGATGTGCTTATTTCACATTCCATGCCTGTATCAAAACATCTCATG
TACCCCATACATGTATATACCTACTATGTATGCATAAAAATTTTTTTAAATAATT
TTTAAAATTAAAAAAATTCTGAGAATTTAAGCTTTTAATCTCAGTTGAGTACTAA
GTGAATGGTAATTAAAGACTAAAATTACTACCAGTTGTCATATTTTAATTGAATT
TAAAATTTACCATTGCAATCATAGAAGTCAACTCTTATTGAAAGCTTAGTCTGTG
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TGAAGCTGCTGAAACCCAAGGTCCCACTACTGGGTGGTGGCAACACTATGAGCTG
TTGGGAAAATATAACTAAATCTATTATTAAGTTTAAGTAAAATAATTGAATATTA
TATAATTATATGAATTAATCAAATATAATTAAATATATTTCATAAATAAATATTA
TTAGAAATAATTCAAGGAGAAAAAAAATCCCCCAACCCAGAAAAACCTGAGG
```

**substrings/kmers of length 8** →

```
'TTTTTTTT', 19
'AAAAAAAT', 3
'AAATAATT', 3
'TAATTTTT', 3
'TATAATTA', 2
'ATATAATT', 2
'AATATTAT', 2
'TATTATTA', 2
'AAATATAA', 2
'AAAATAAT', 2
'AAAATGTA', 2
'TAAAATGT', 2
'CTAAAATG', 2
'TCTAAAAT', 2
'AATTGAAT', 2
'TAATTGAA', 2
'TAATTAAA`, 2
...
```

**Identify what is**

- **unique**
- **frequent**
- **differentially frequent**

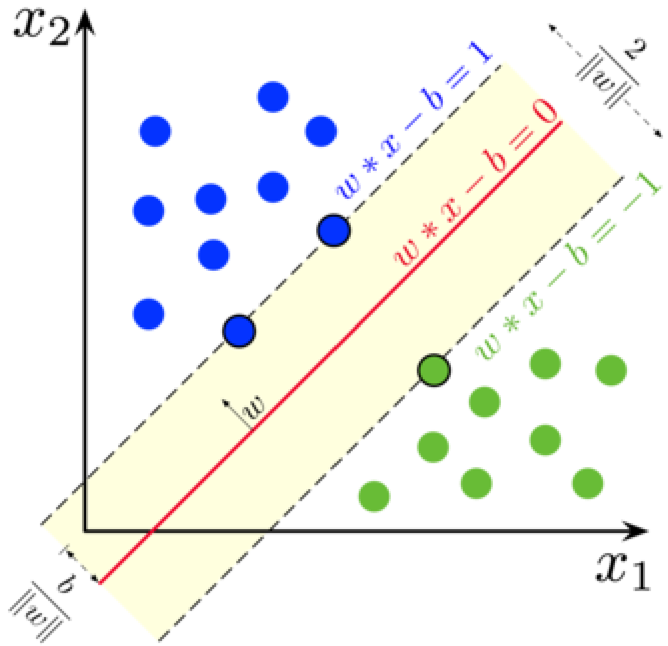
**…**

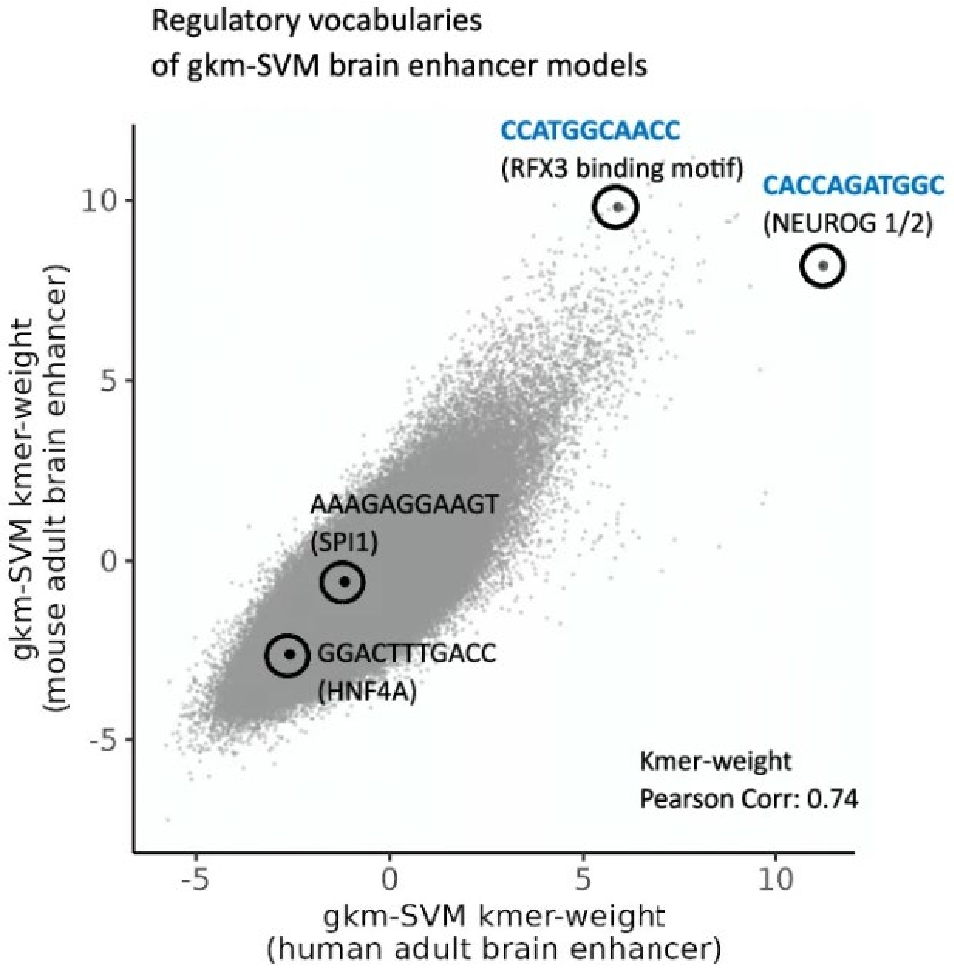
# gapped-kmer SVMs

**Gaps as „no information“ positions**

GGAATTGTAG -G--TTGTAG

CGGTTTGTAG -G--TTGTAG




https://www.nature.com/articles/s41467-024-50708-z/figures/1

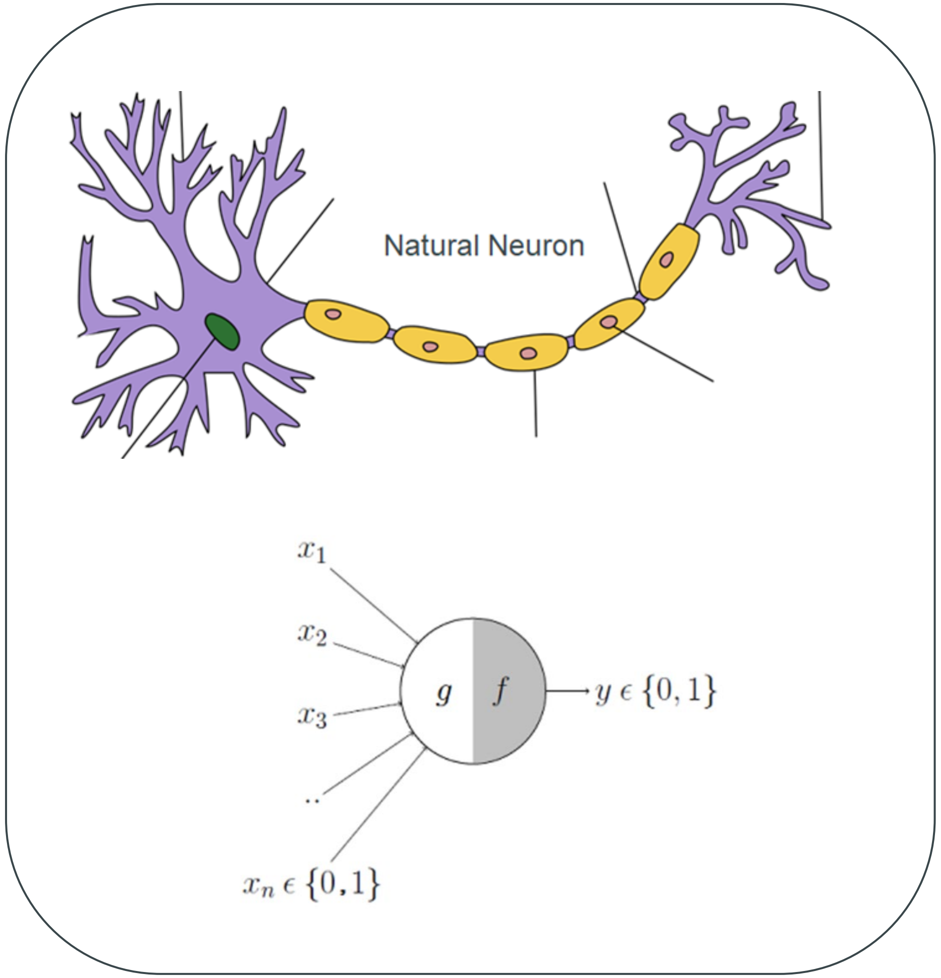
# Neural networks (NNs)

Artificial
Neural
Networks

Born in the 1950s
Died in the 1970s

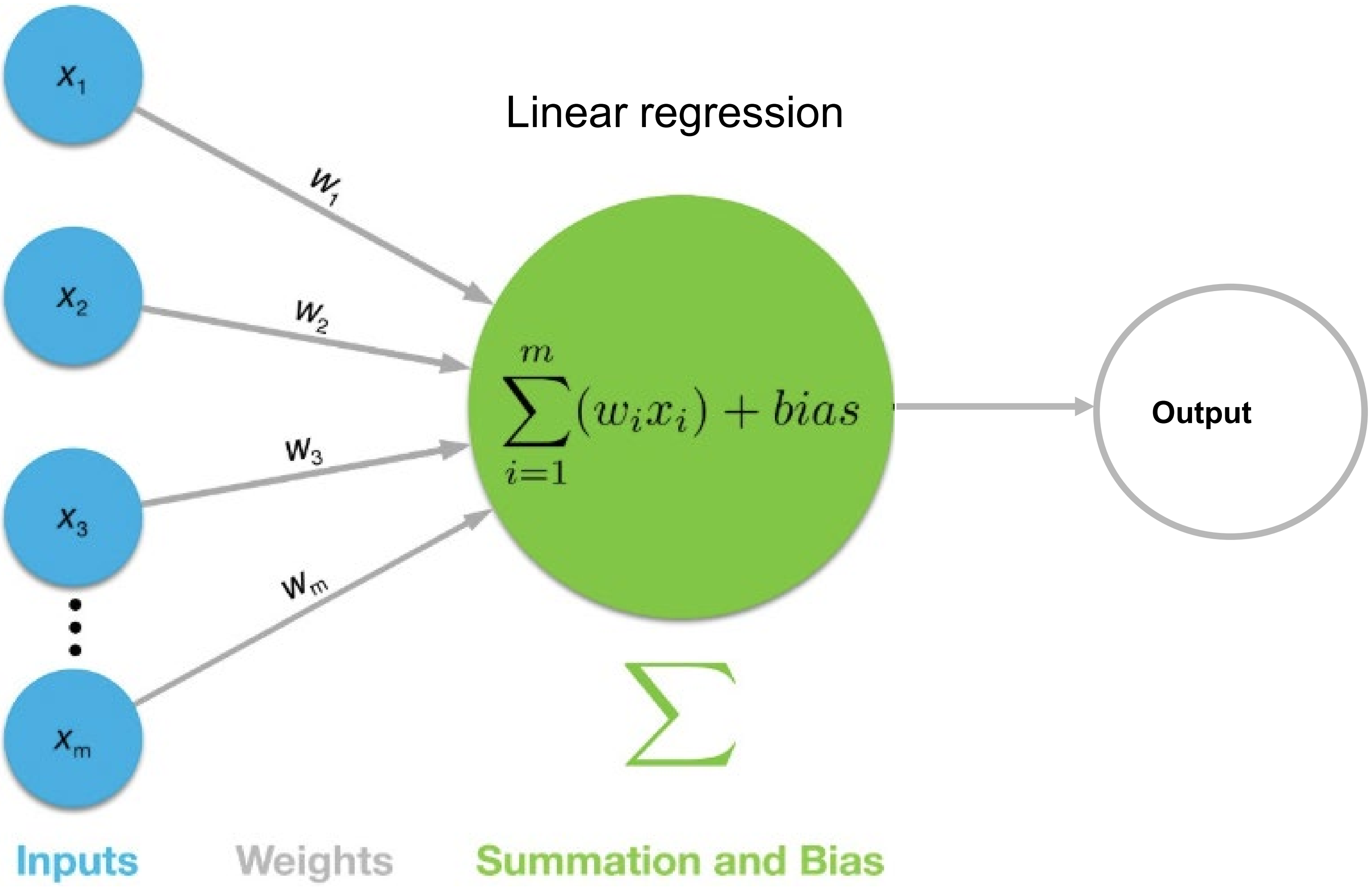
Reborn in 1980s
Died again in the 1990s

Reborn in the 2010s
Currently booming field




Modified from CEITEC ECCB-2022 workshop

# Neural networks (NNs)

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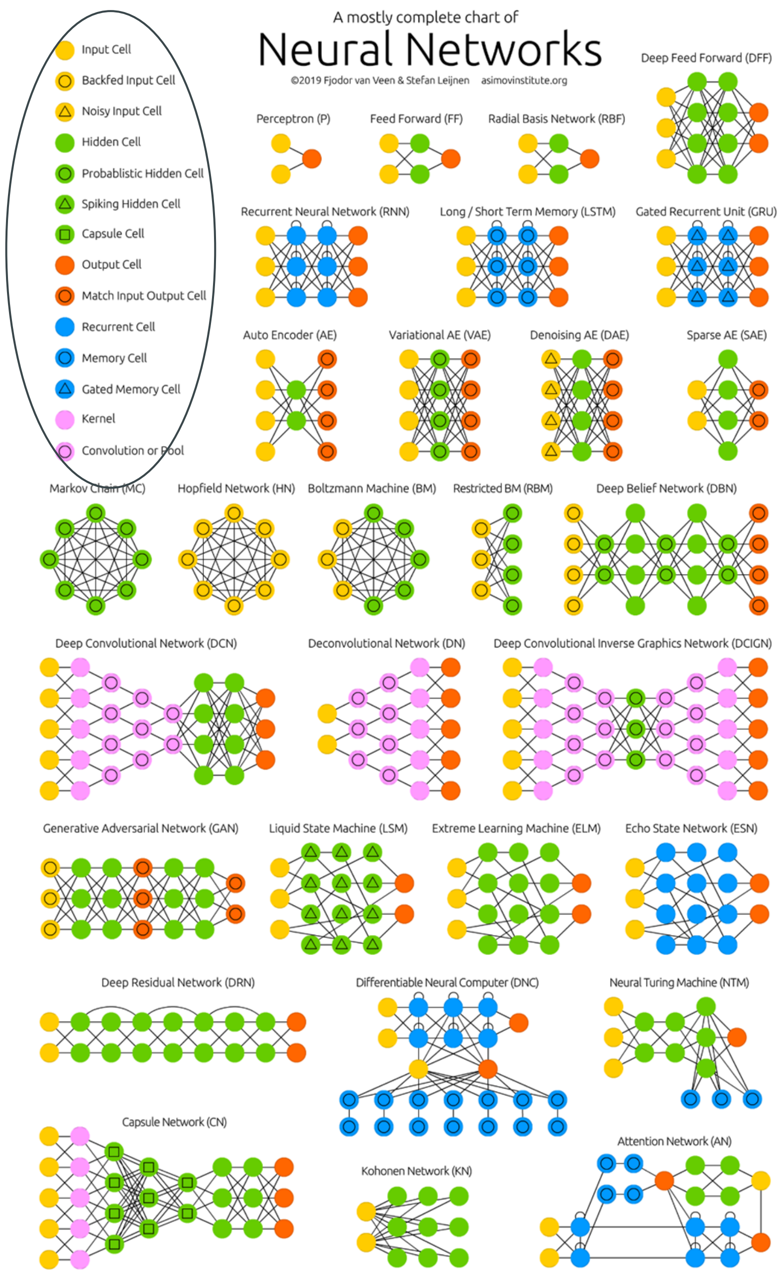
# Deep Neural networks (NNs)

simple
building
blocks

neurons




**Deep Neural Networks**

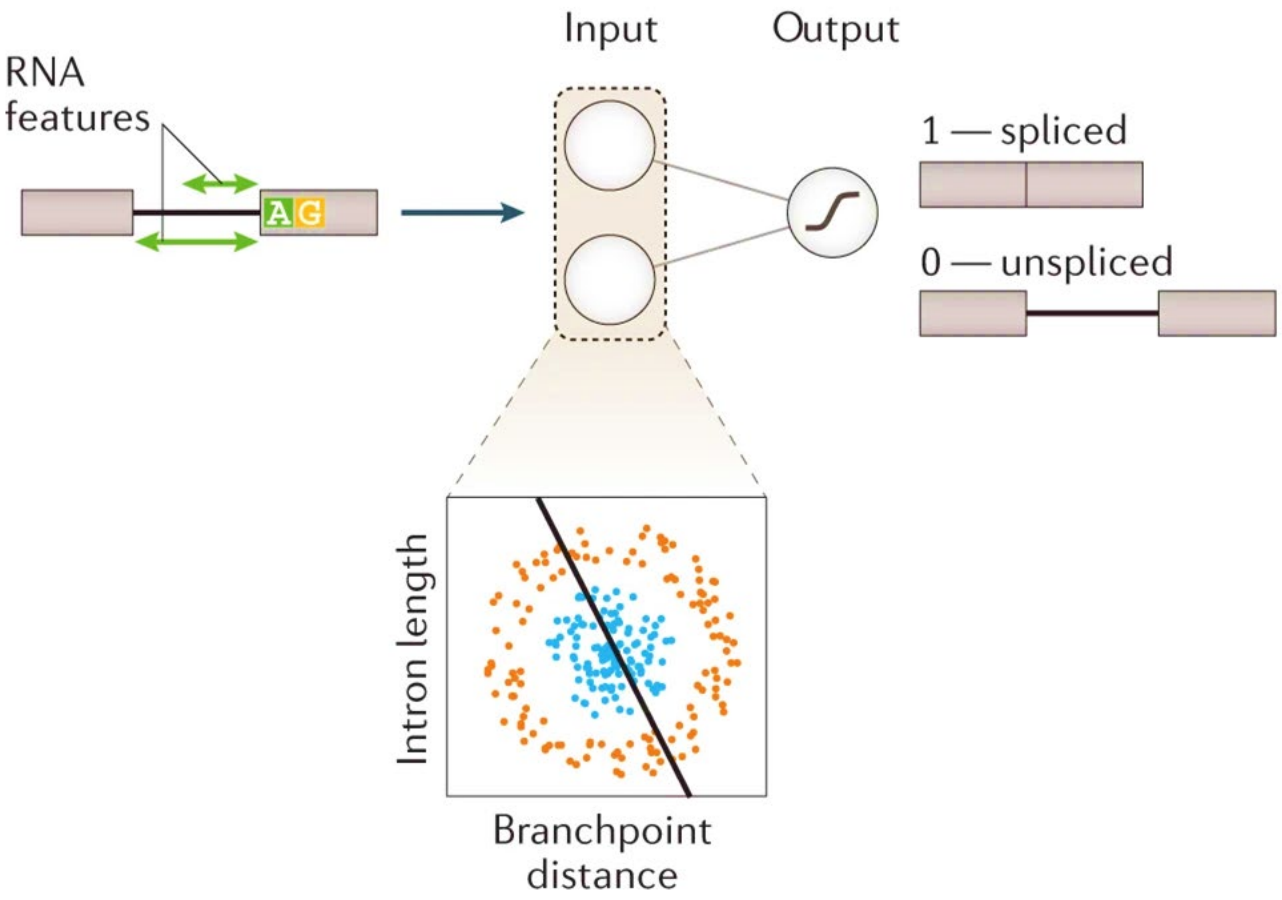
use **several layers** of neurons
to model increasingly **complex**
patterns in data.

Modified from CEITEC ECCB-2022 workshop



12

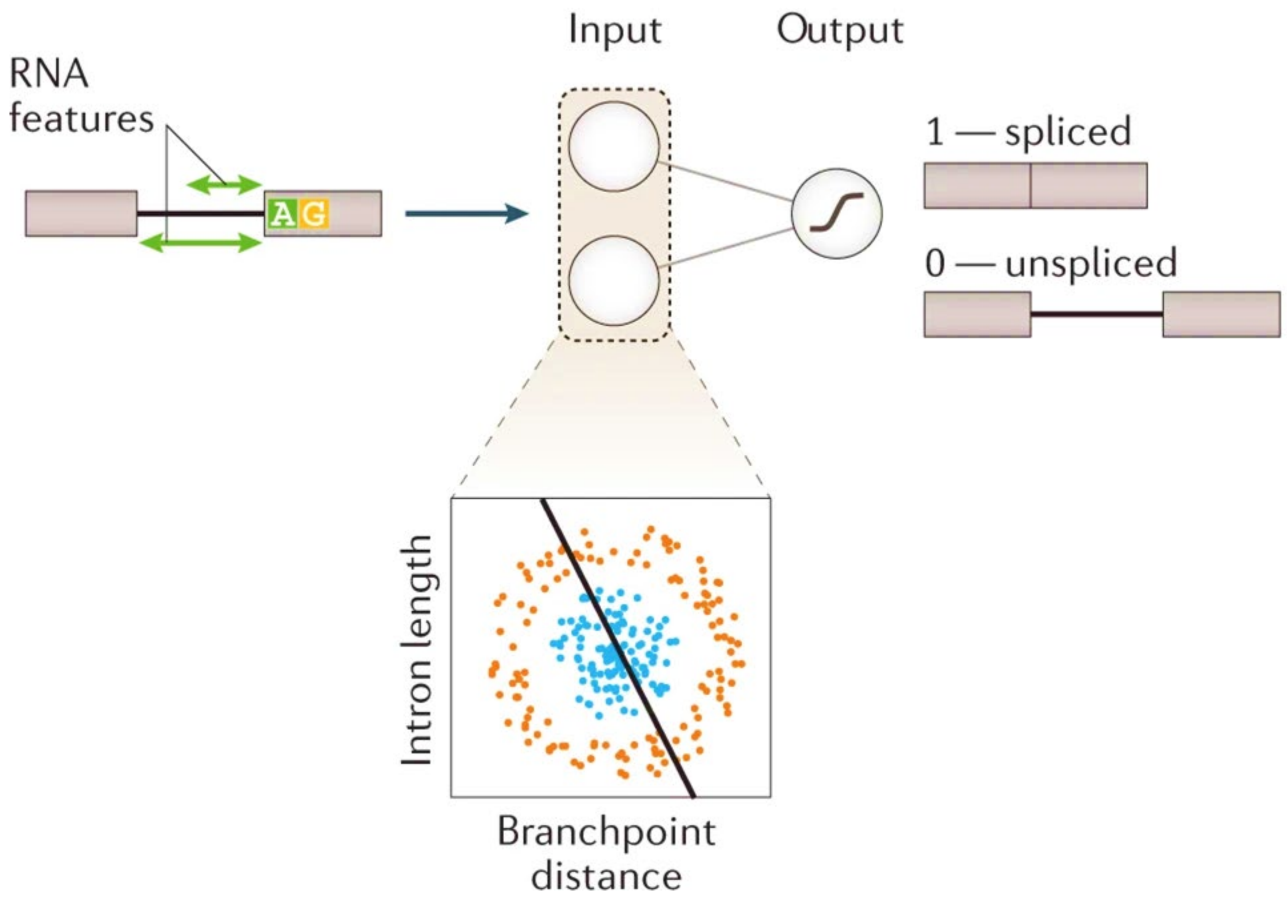
# Why neural networks? (NNs)

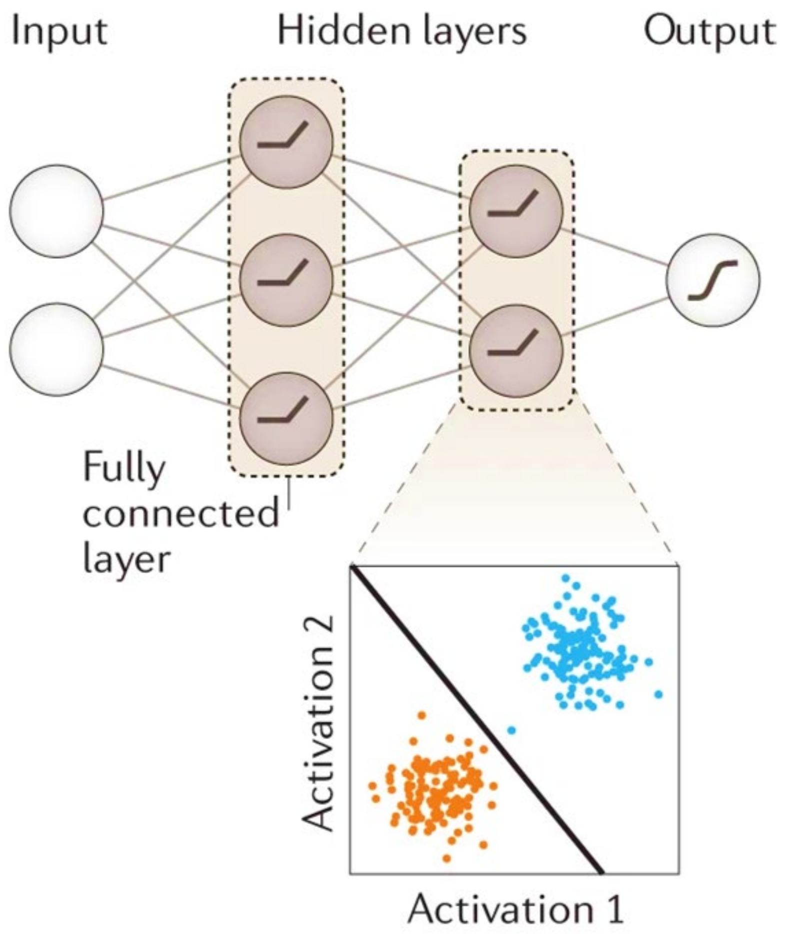





Eraslan, G., Avsec, Ž., Gagneur, J. et al. Deep learning: new computational modelling techniques for genomics. Nat Rev Genet 20, 389–403 (2019).

# Enter Deep Neural Networks (DNNs)



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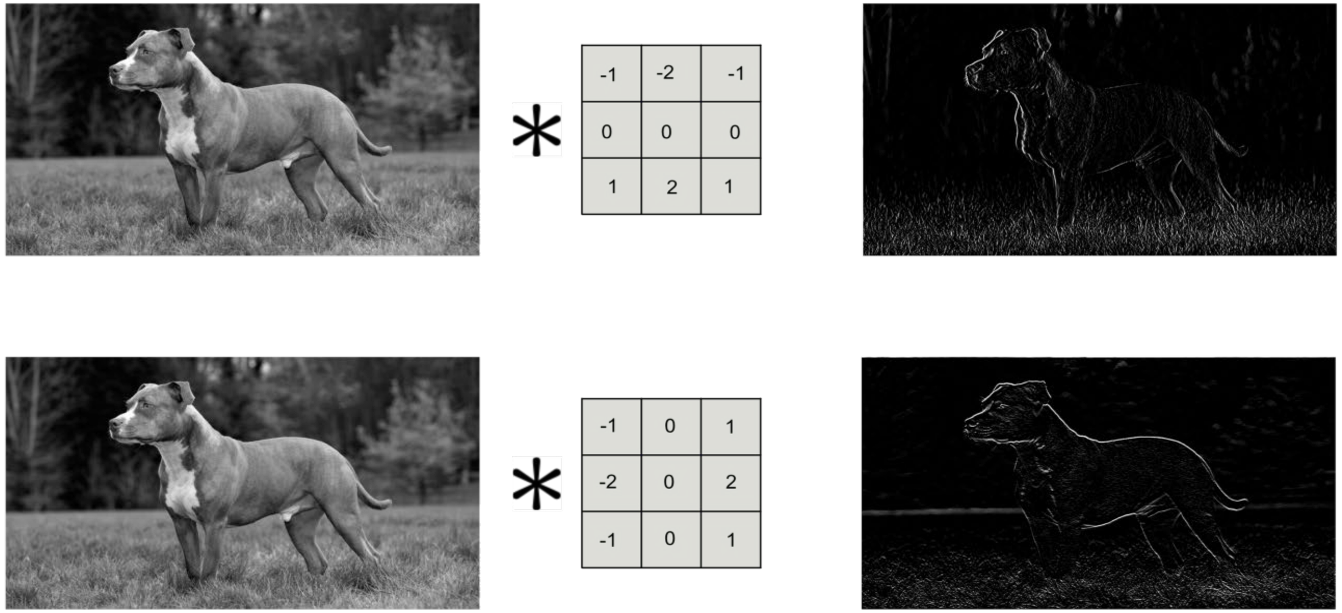
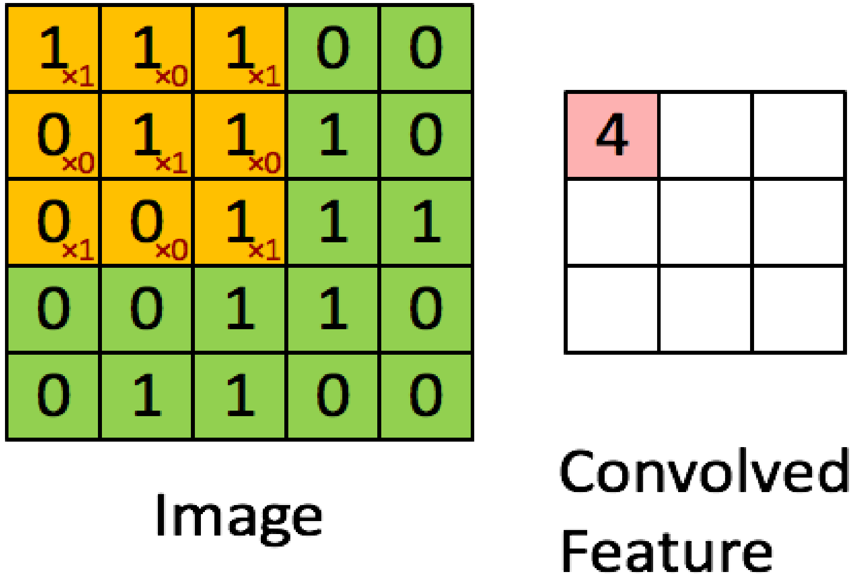
# Convolutional Neural networks (CNNs)

Looks for patterns in input (for e.g., an image), independently of their location

Matrix defines the pattern (filter)

Multiplying the image -> number representing the ‘fit’

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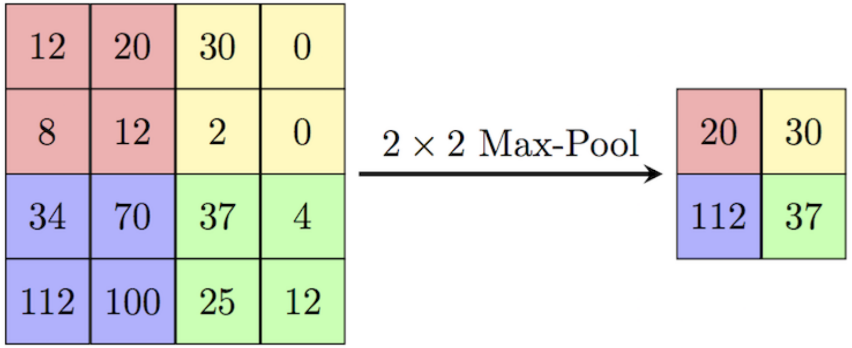
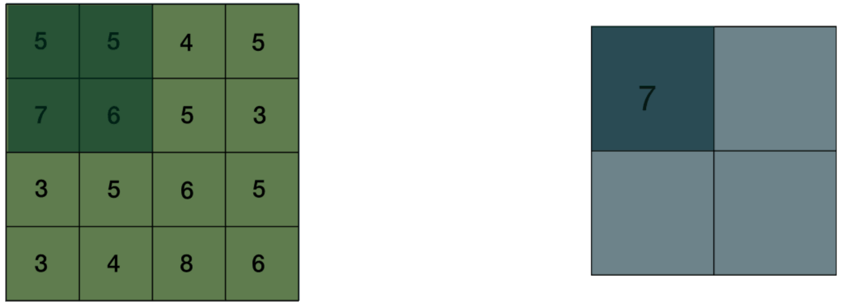
# Pooling layer

(Max.) - Pooling - taking the most significant areas

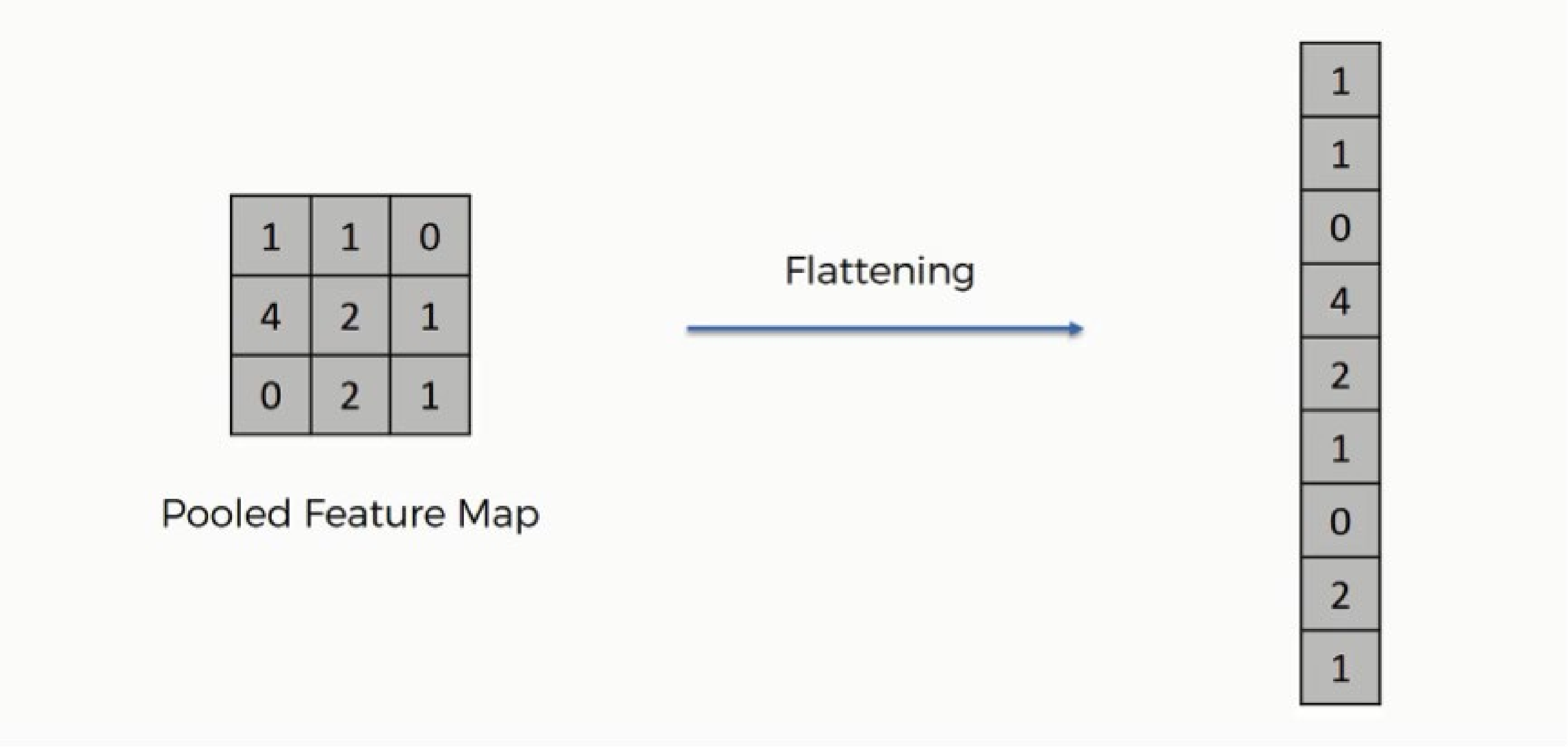



Modified from CEITEC ECCB-2022 workshop

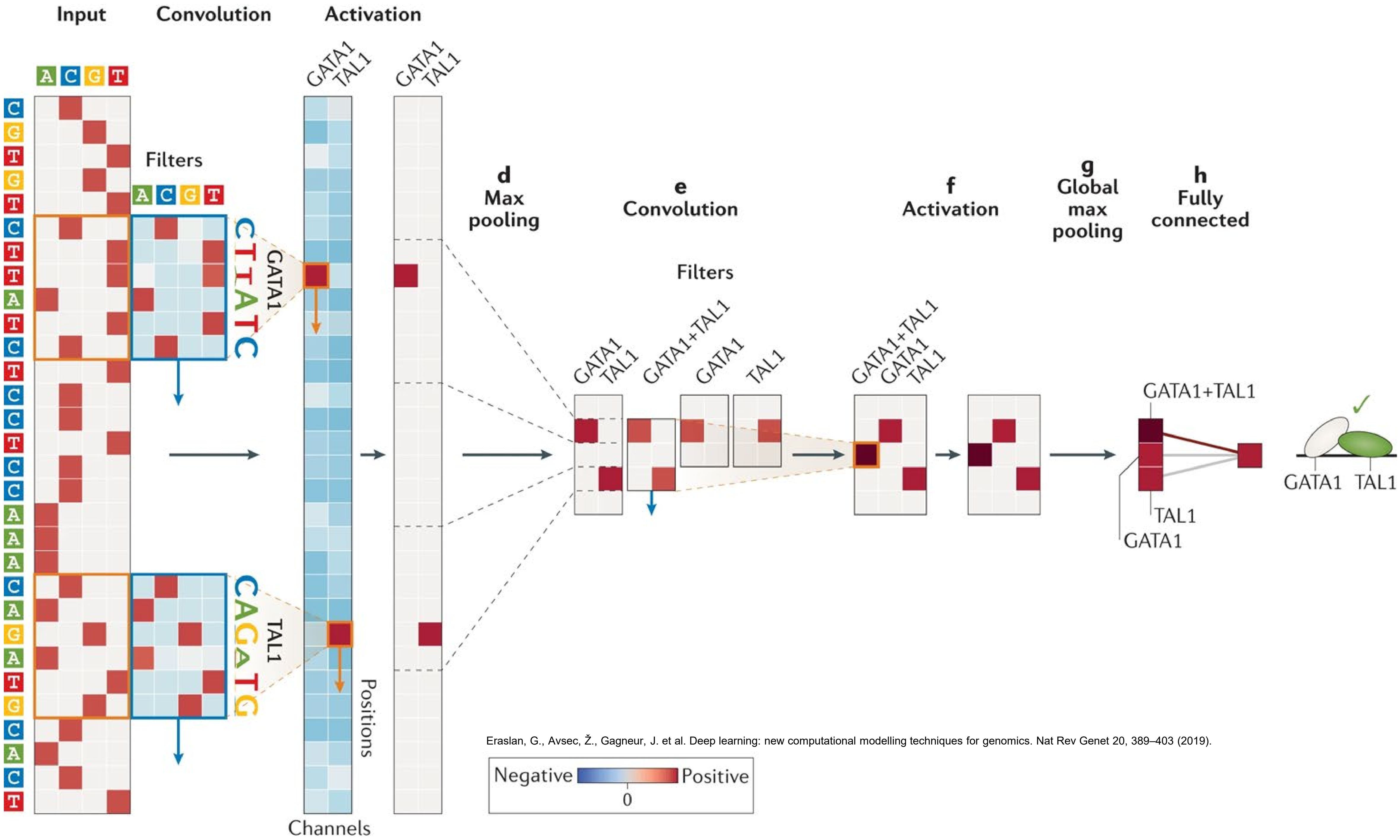
# Flatten layer

Flatten - Spread the pixels and then take combinations of these






Modified from CEITEC ECCB-2022 workshop

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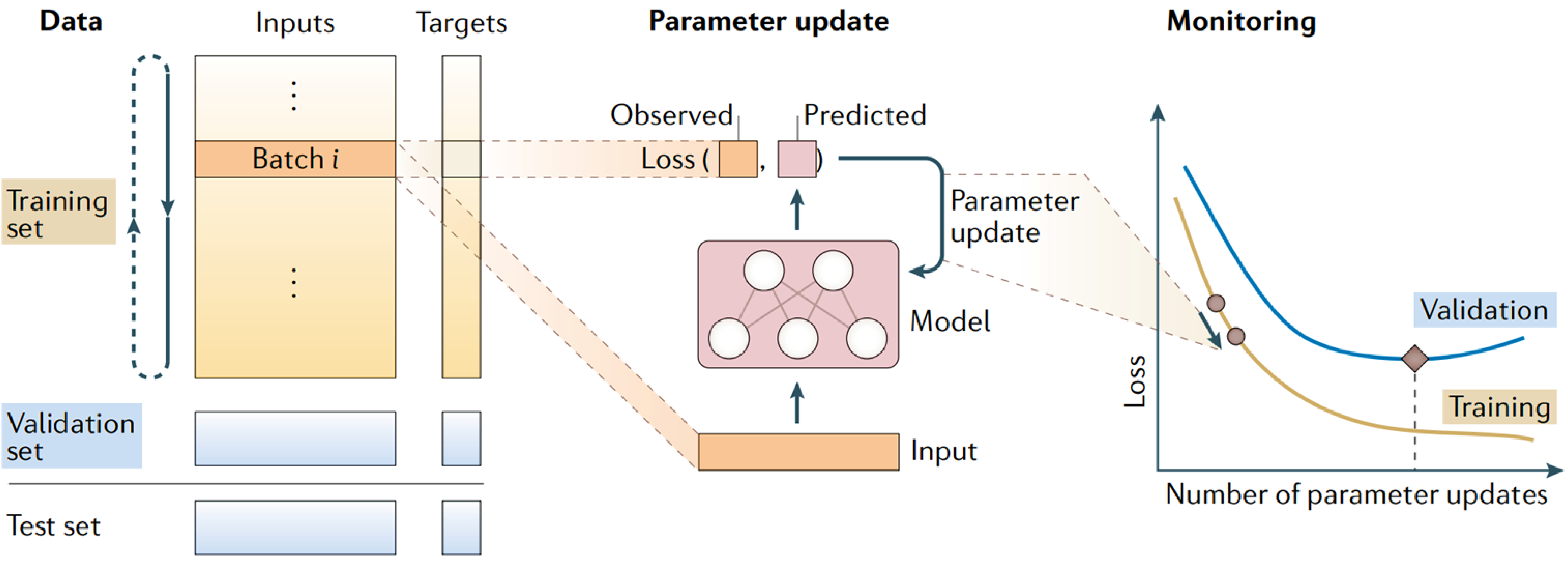
# Model training

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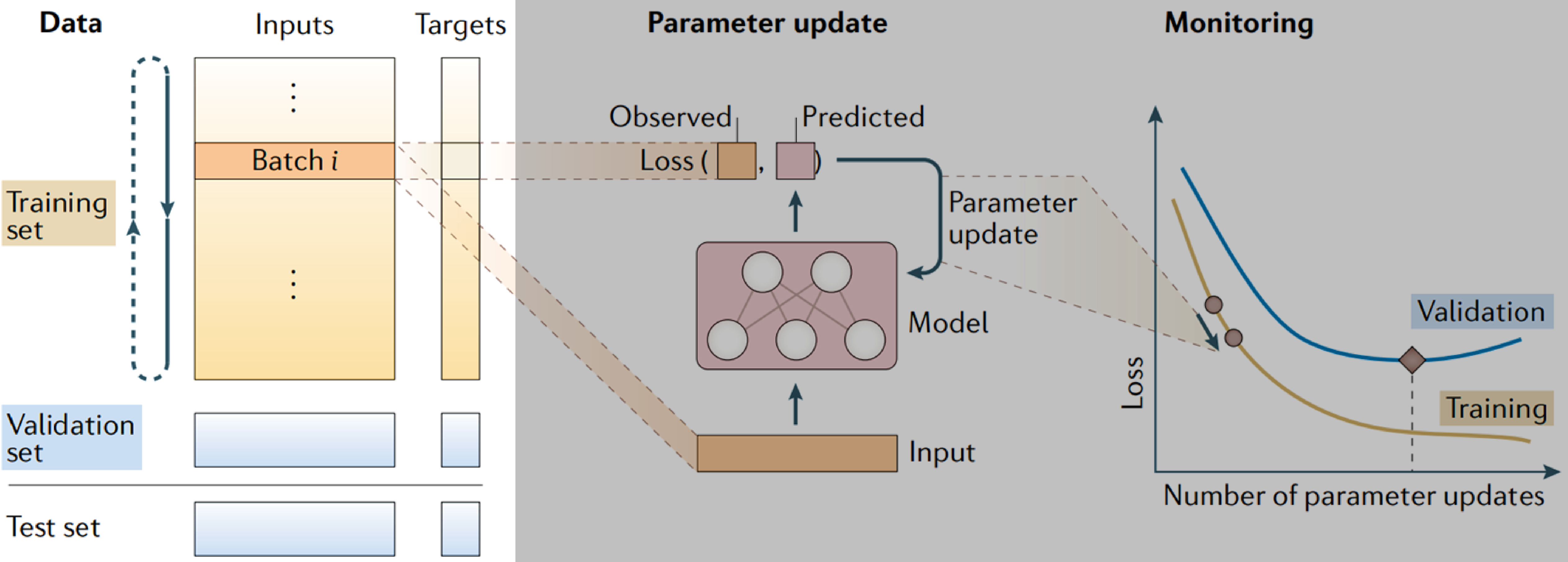
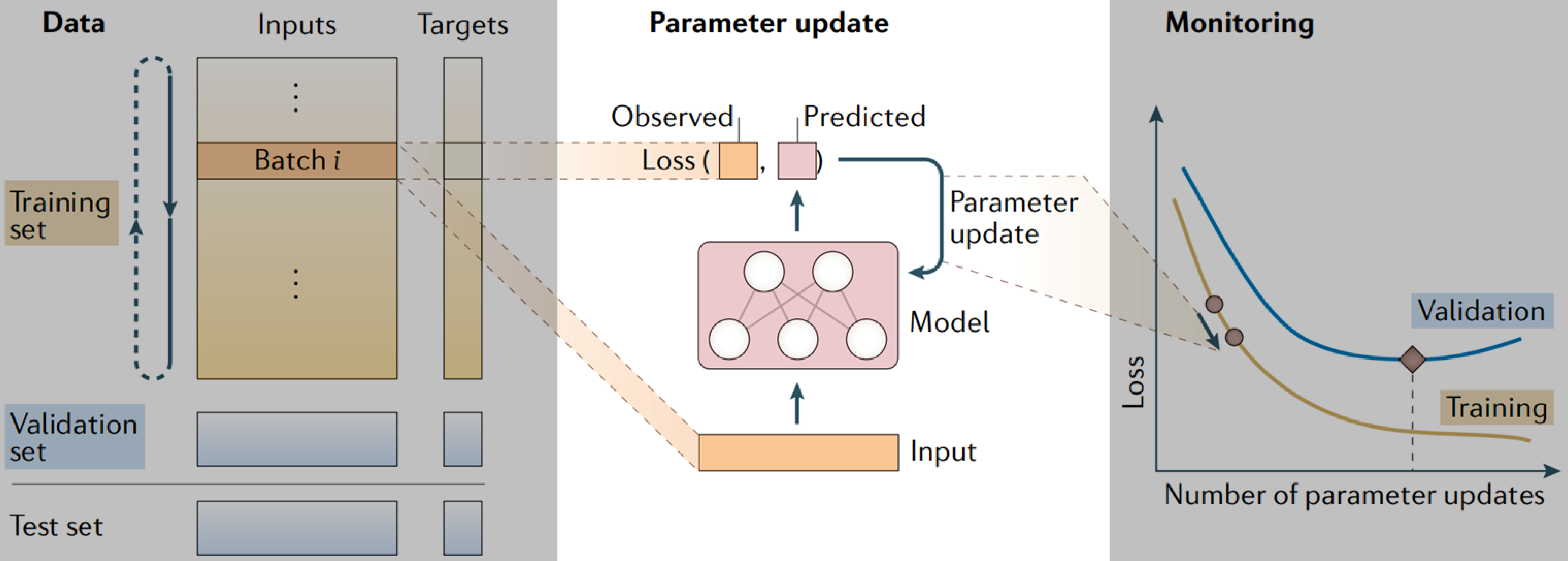
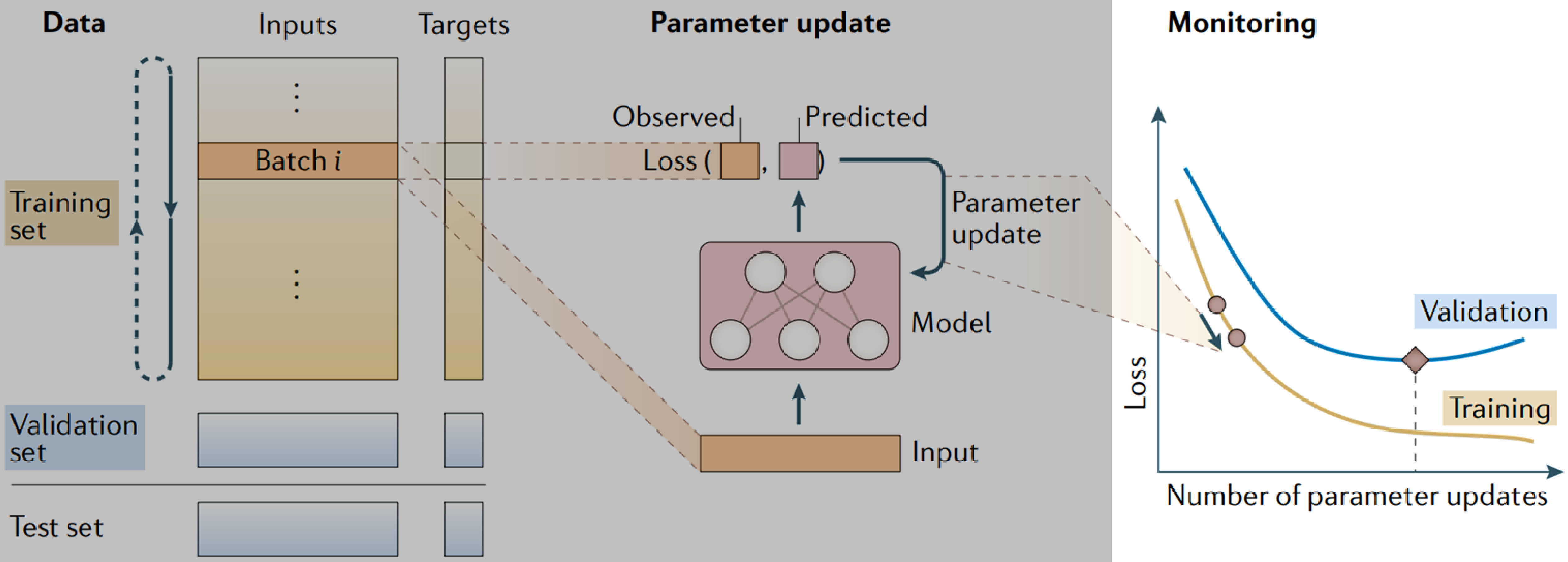



Data
Inputs
Targets
Batch i
Training set
Validation set
Test set
Parameter update
Observed
Predicted
Loss ( , )
Parameter update
Model
Input
Monitoring
Loss
Validation
Training
Number of parameter updates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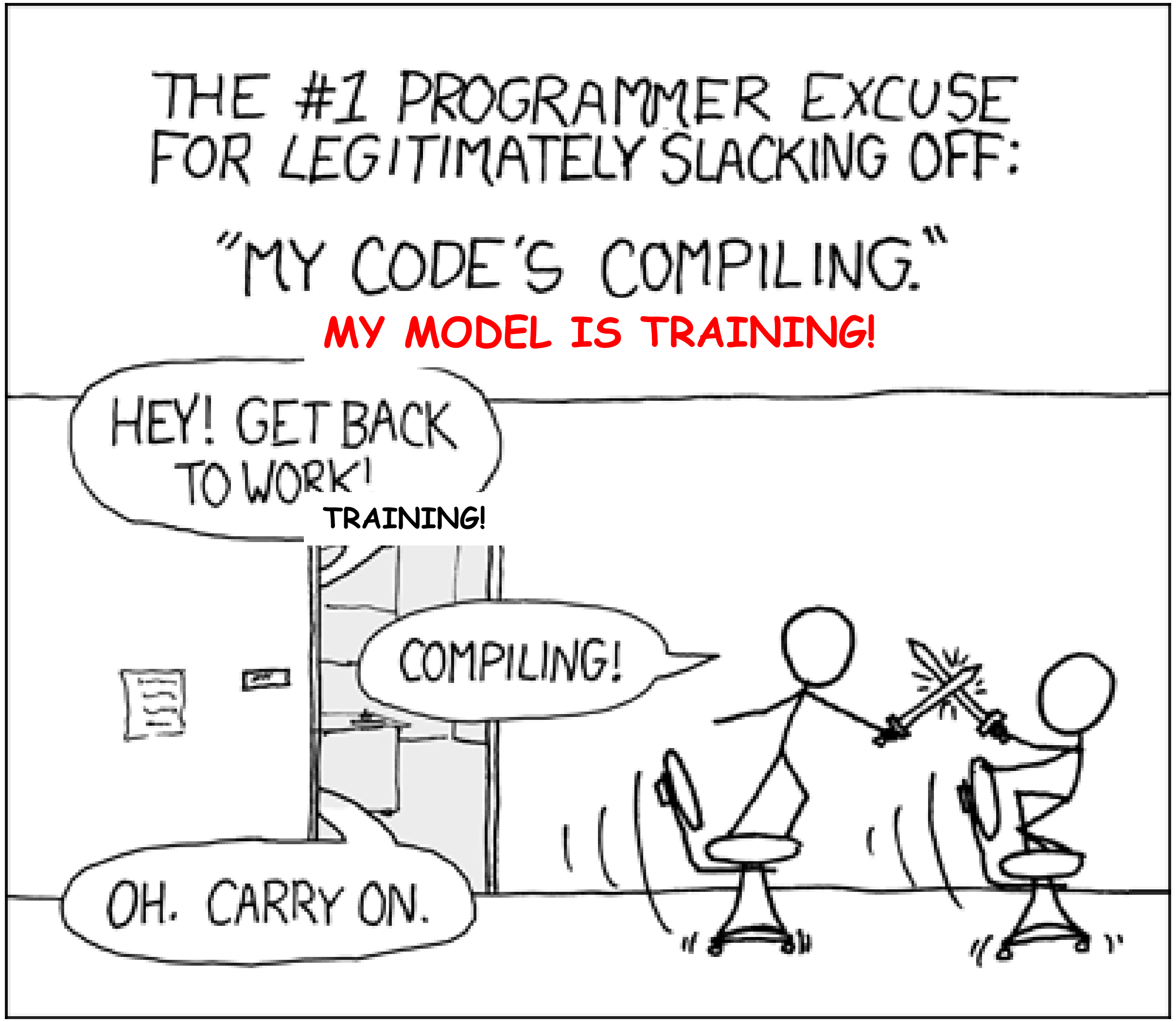
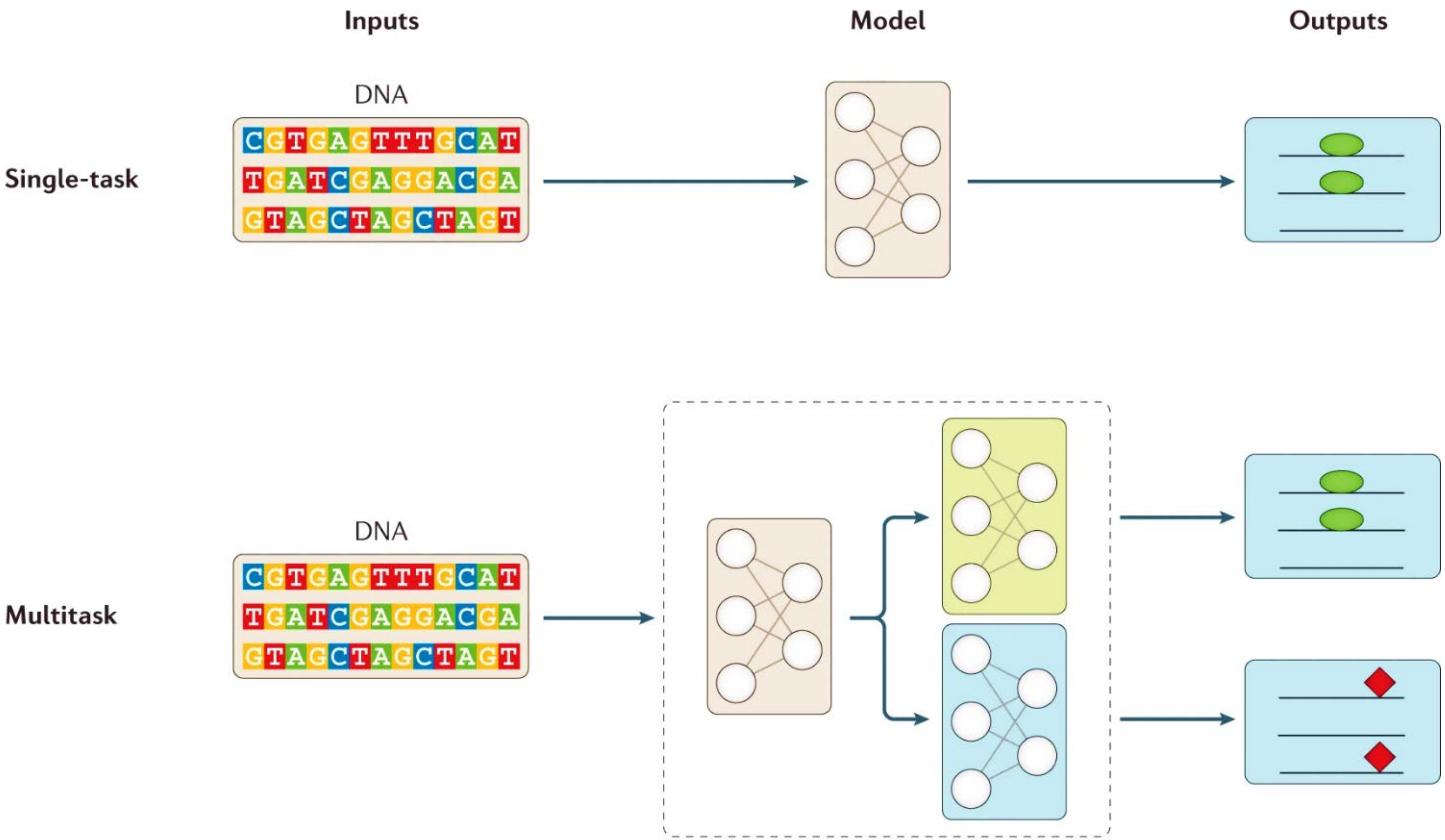
Monitoring
Loss
Validation
Training
Number of parameter updates

# Hmm…




https://xkcd.com/303/

# Model training – How to *stir the pile*?

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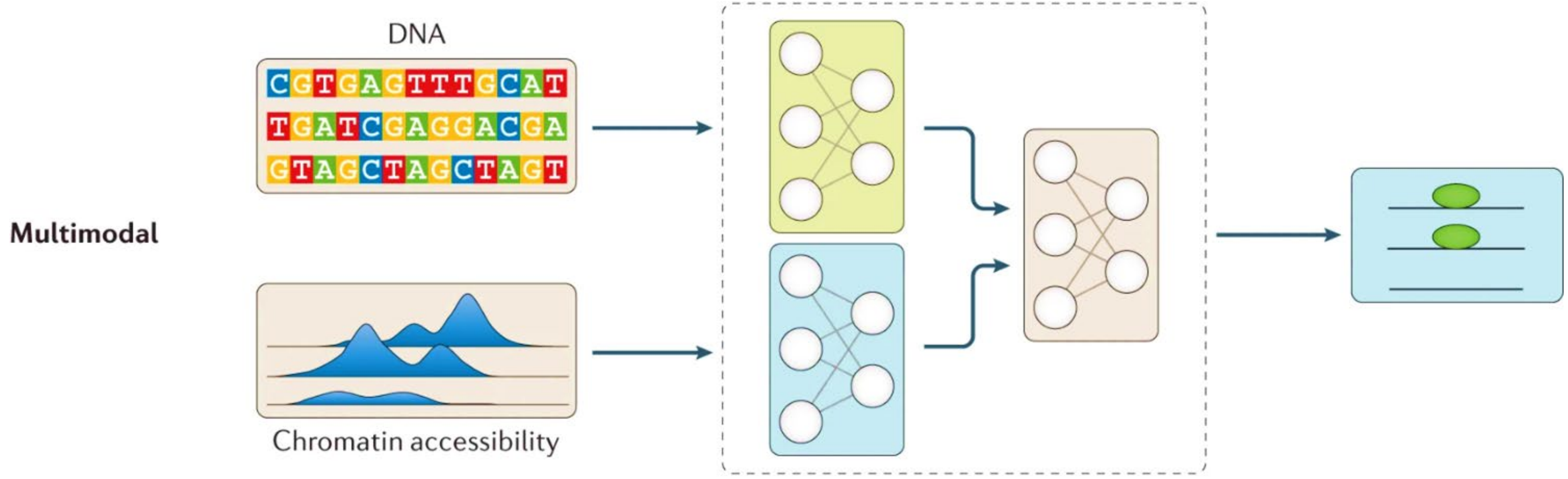




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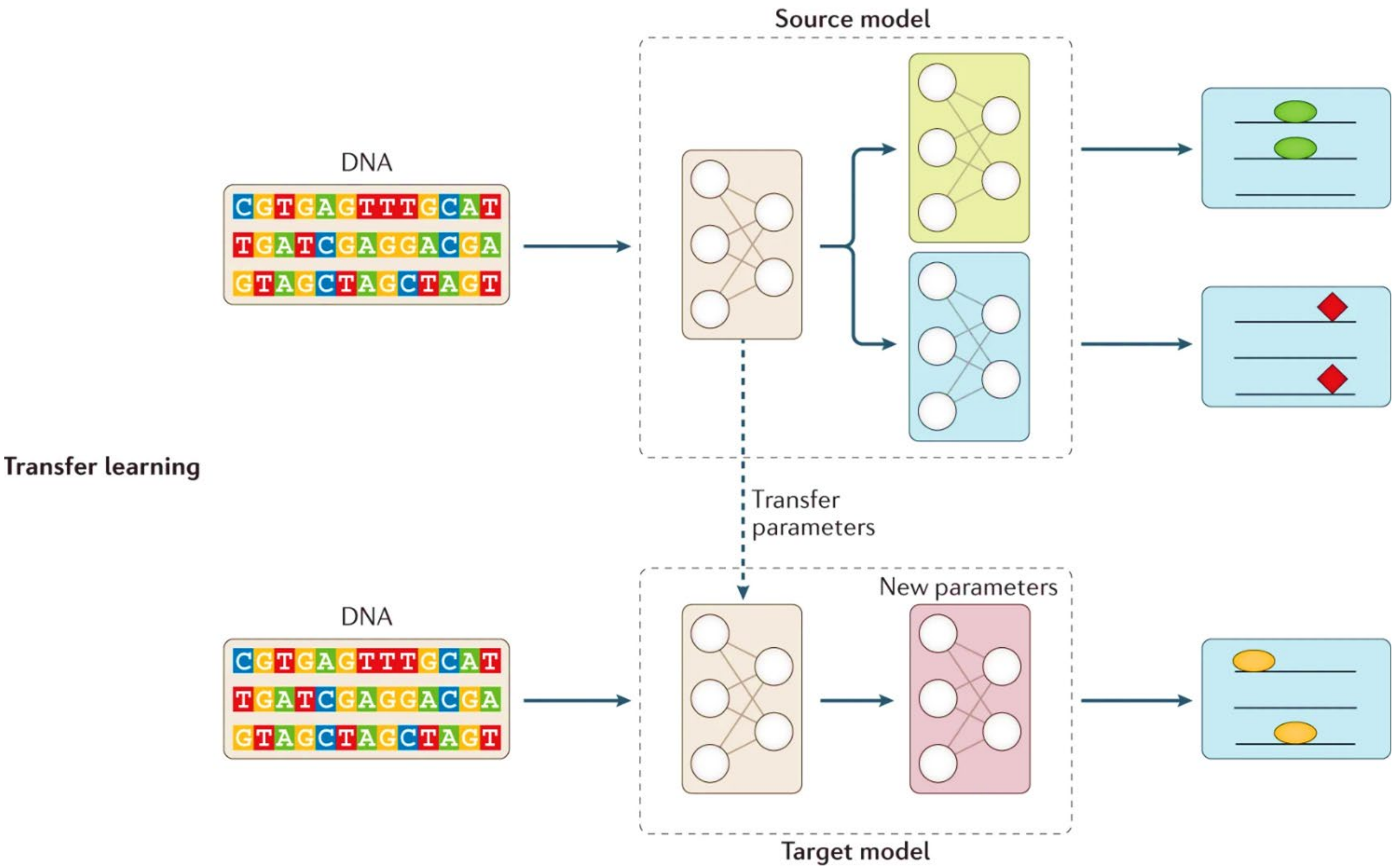





Eraslan, G., Avsec, Ž., Gagneur, J. et al. Deep learning: new computational modelling techniques for genomics. Nat Rev Genet 20, 389–403 (2019).

# Useful resources

Goodfellow, I., Bengio, Y. & Courville, A. *Deep Learning* (MIT Press, 2016). This textbook covers theoretical and practical aspects of deep learning with introductory sections on linear algebra and machine learning.

Eraslan, G., Avsec, Ž., Gagneur, J. et al. Deep learning: new computational modelling techniques for genomics. Nat Rev Genet 20, 389–403 (2019).

https://huggingface.co/
The AI community building the future.

https://www.3blue1brown.com/

3Blue1Brown

Kipoi https://kipoi.org/

Model Zoo for genomics

# Useful resources (2)

https://www.kaggle.com/

Center for Critical Assessment of Genome Interpretation   **CAGI Scientist Opening:** apply here   Register/Login

## Critical Assessment of Genome Interpretation

Challenges   Meetings   Publications   Organization   Contact Us

https://genomeinterpretation.org/

## Challenges

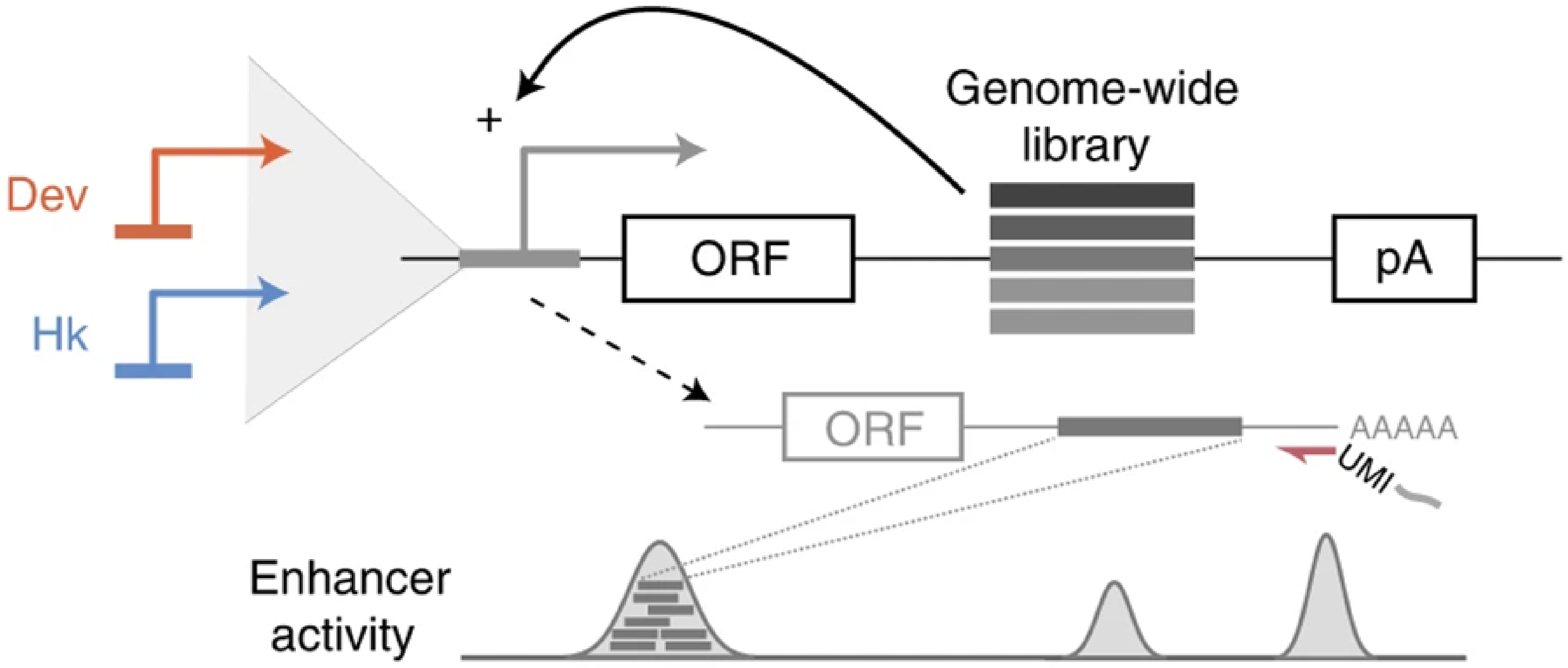
https://paperswithcode.com/

SageBionetworks   About   How We Partner   Tools & Resources   Careers   News   Donate

https://sagebionetworks.org/tools_resources/challenges/

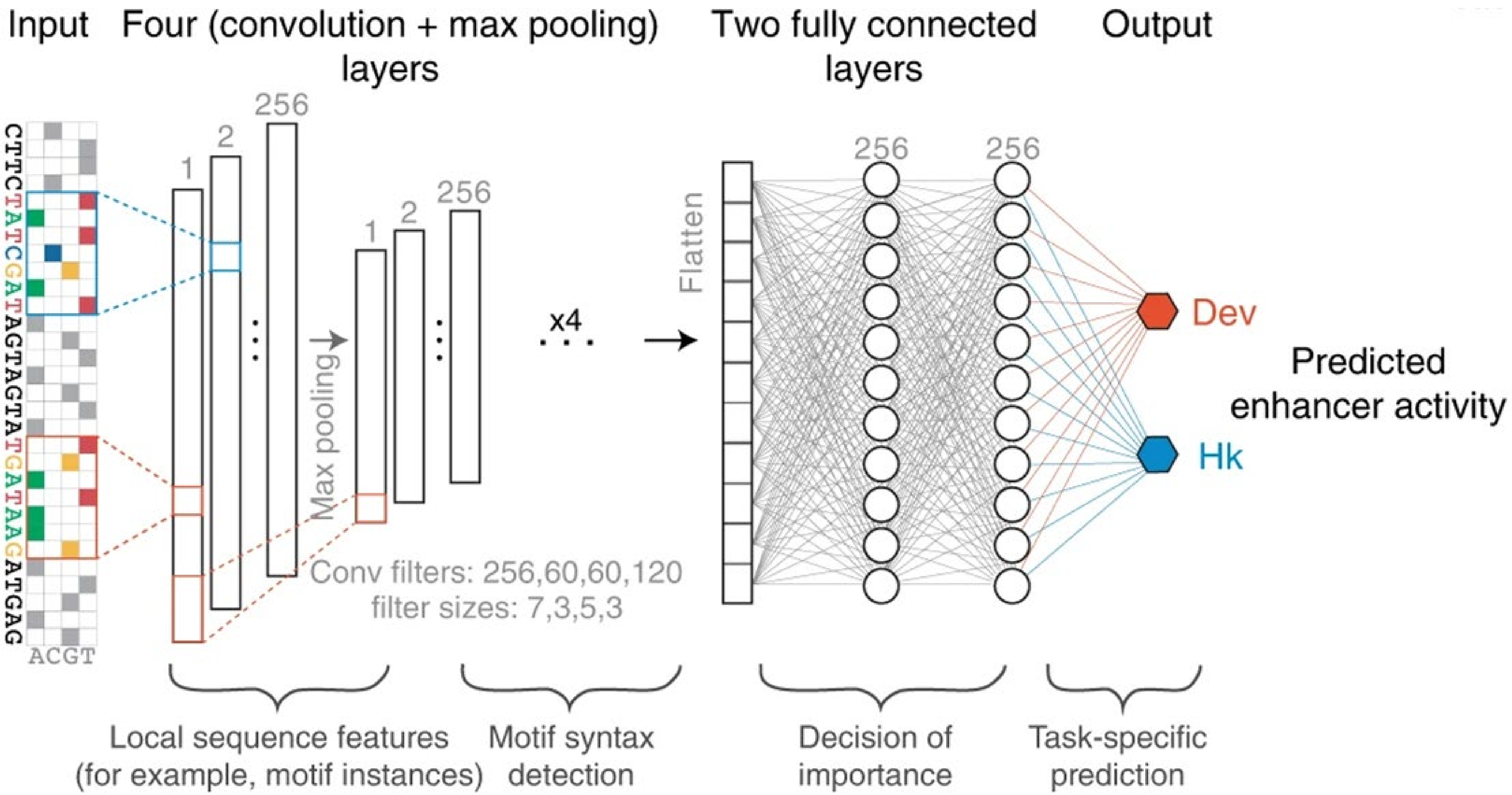
Challenge Platform

# Example: DeepSTARR




de Almeida, B.P., Reiter, F., Pagani, M. et al. DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. Nat Genet 54, 613–624 (2022).

# Example: DeepSTARR

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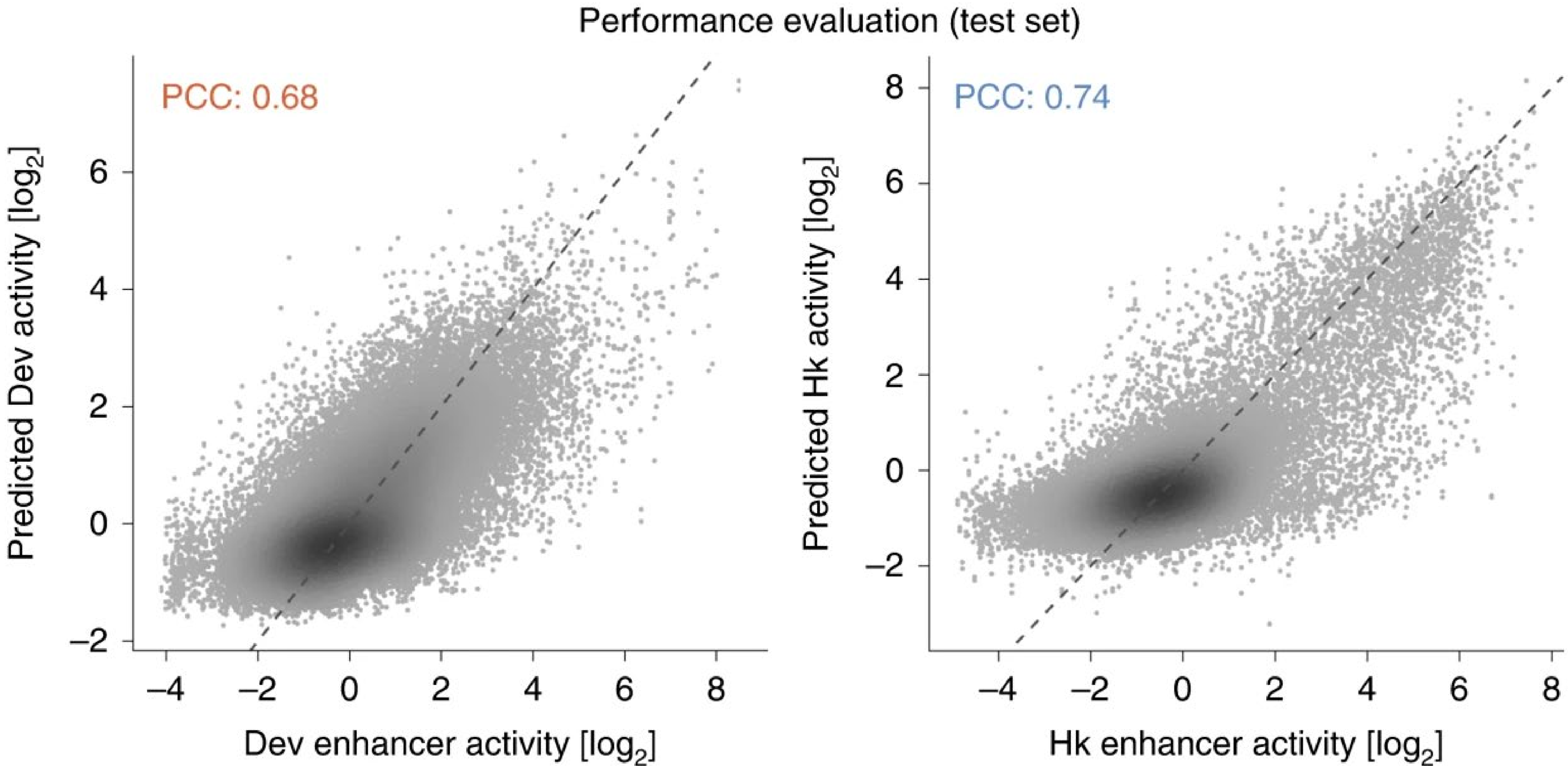
# Example: DeepSTARR

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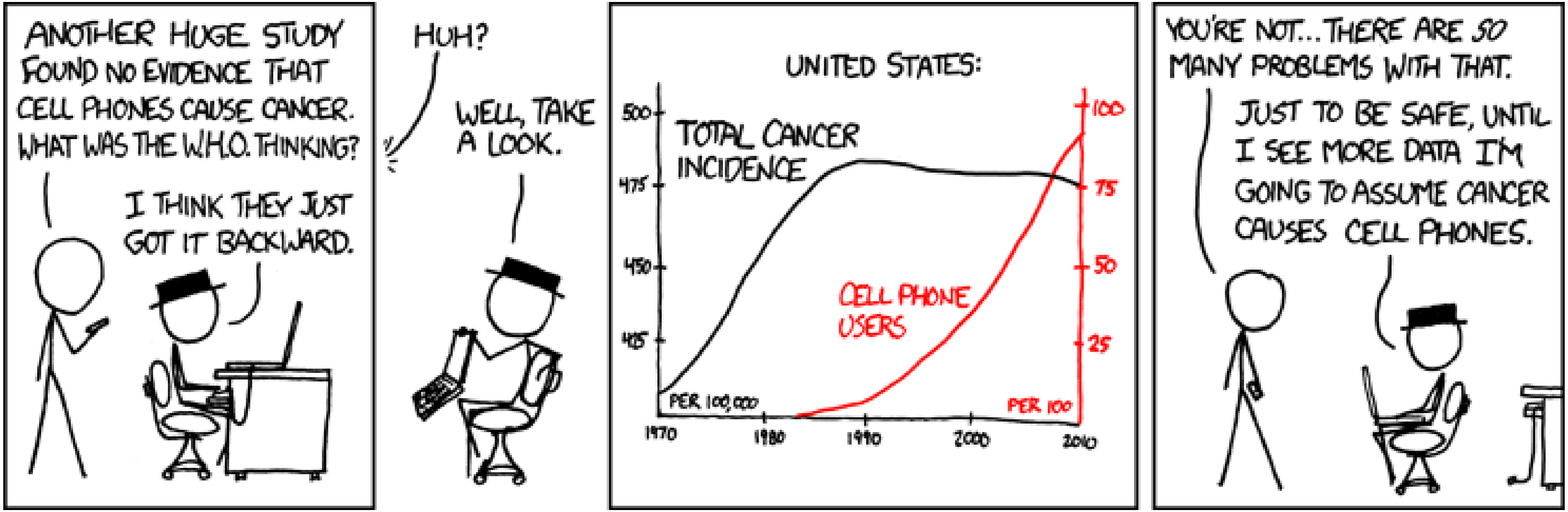
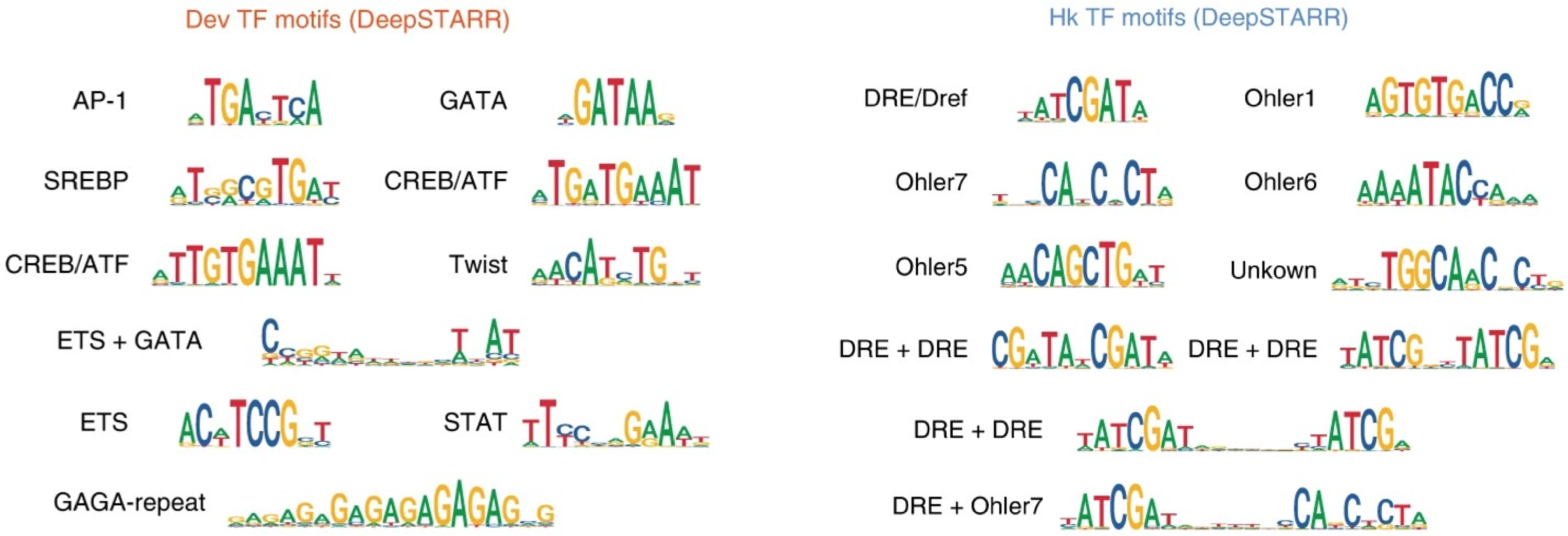
# Model interpretation



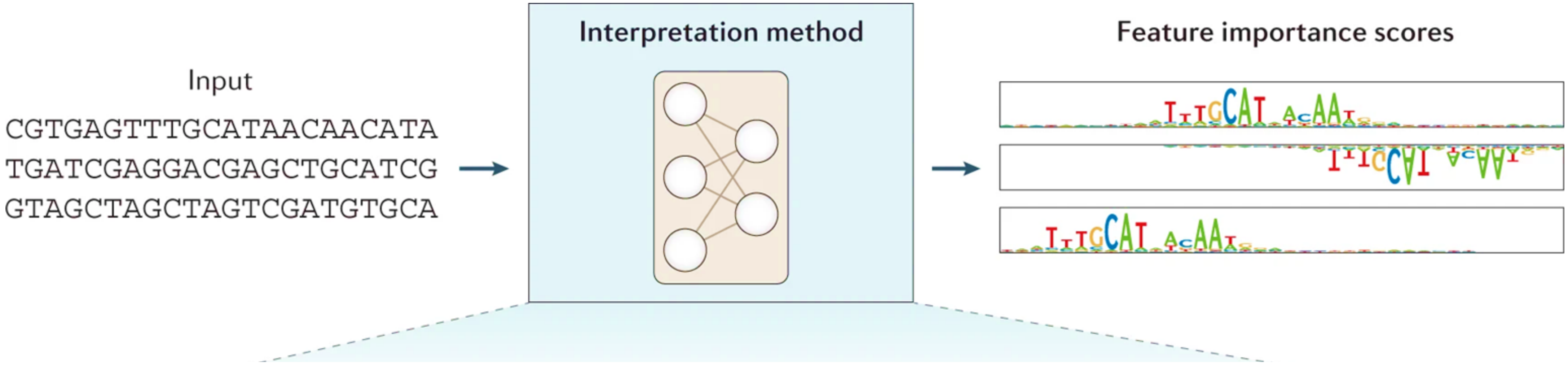

https://xkcd.com/925/

# Example: DeepSTARR




de Almeida, B.P., Reiter, F., Pagani, M. et al. DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. Nat Genet 54, 613–624 (2022).

# Model Interpretation - the idea



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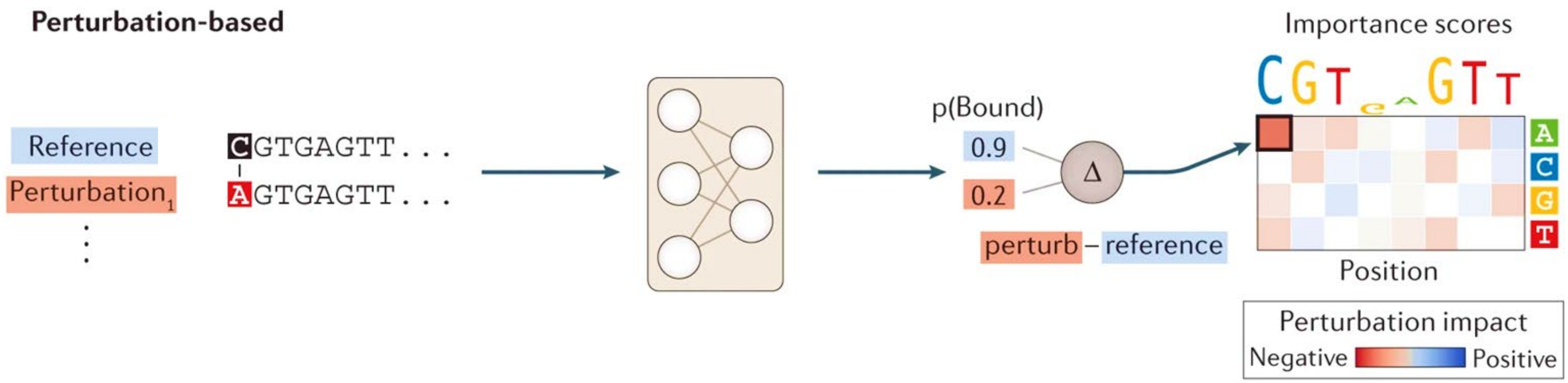
# Model Interpretation - ISM






Eraslan, G., Avsec, Ž., Gagneur, J. et al. Deep learning: new computational modelling techniques for genomics. Nat Rev Genet 20, 389–403 (2019).

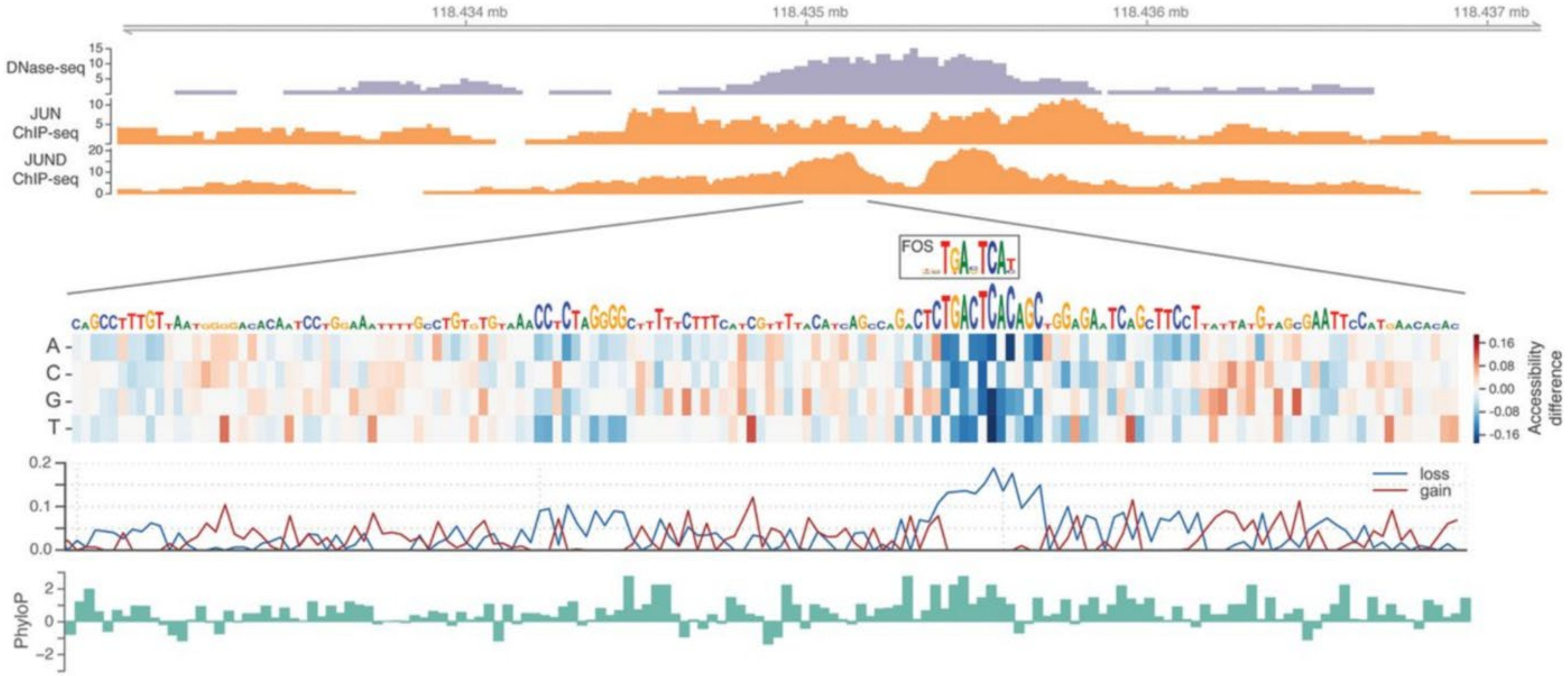
# Model Interpretation - ISM

**Basset: learning the regulatory code of the accessible genome with deep convolutional neural networks**

David R. Kelley[1], Jasper Snoek[2] and John L. Rinn[1]

+ Author Affiliations

Corresponding author: dkelley@fas.harvard.edu



# Acknowledgements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- Thorben Maaß (UKSH – Lübeck, Germany)




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kircherlab.github.io

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UK SH UNIVERSITÄTSKLINIKUM Schleswig-Holstein

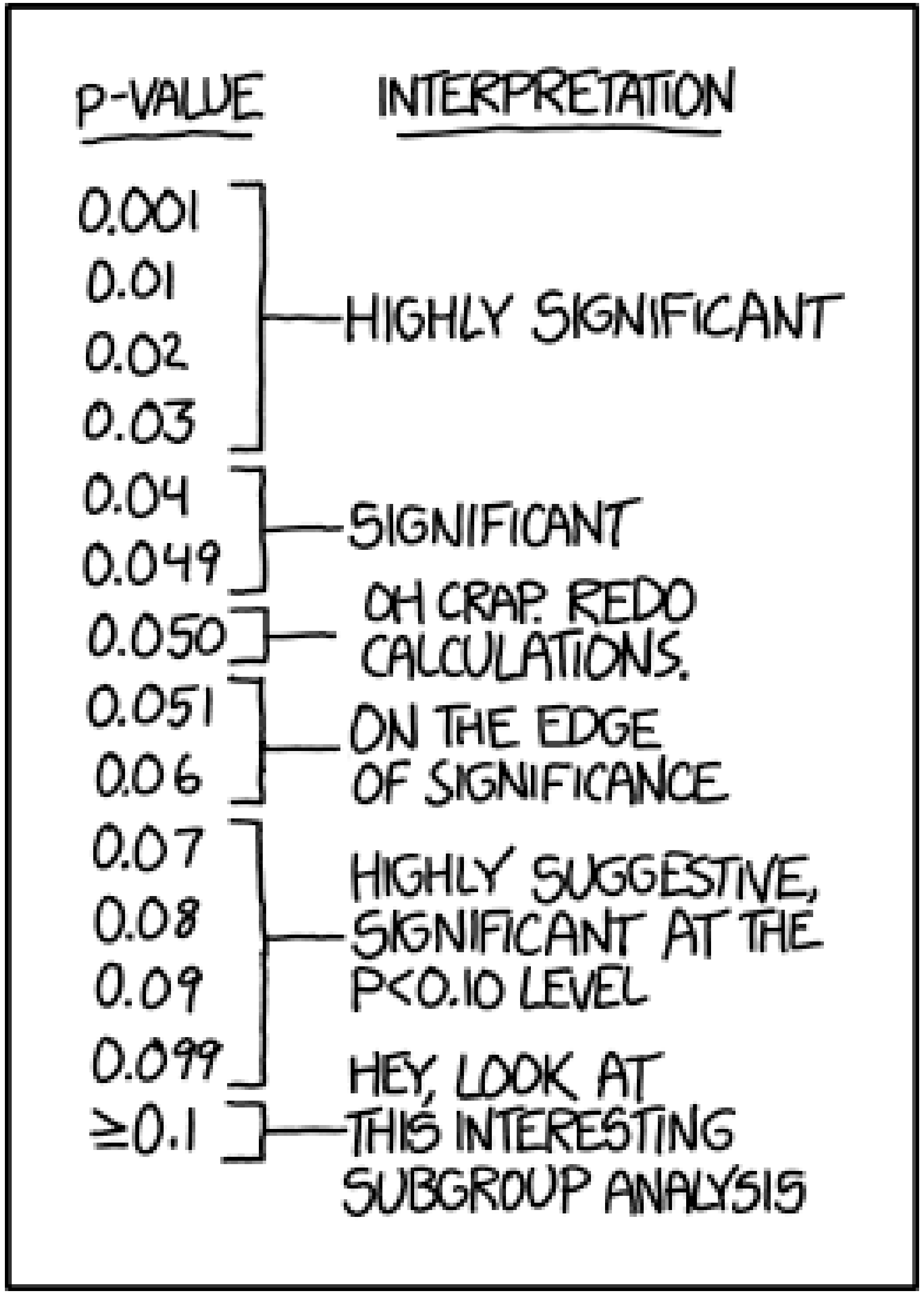
BIH Berlin Institute of Health @Charité

# Hands-On sessions: Model training

**ISMB-ECCB 2025:** *Tutorial IP2 (MPRAs and IGVF Resource)*

**Max Schubach**
**Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH**

July 20, 2025




https://xkcd.com/1478/

# Hands-on: Training a sequence-based model

https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

**Go to:** 05_sequence_models/01_training_sequence_model.ipynb

**Klick:** Open in Colab

# Discussion of pre-training and fine-tuning approaches

**ISMB-ECCB 2025:** *Tutorial IP2 (MPRAs and IGVF Resource)*

**Arjun Devadas, Kilian Salomon, Max Schubach, Martin Kircher**
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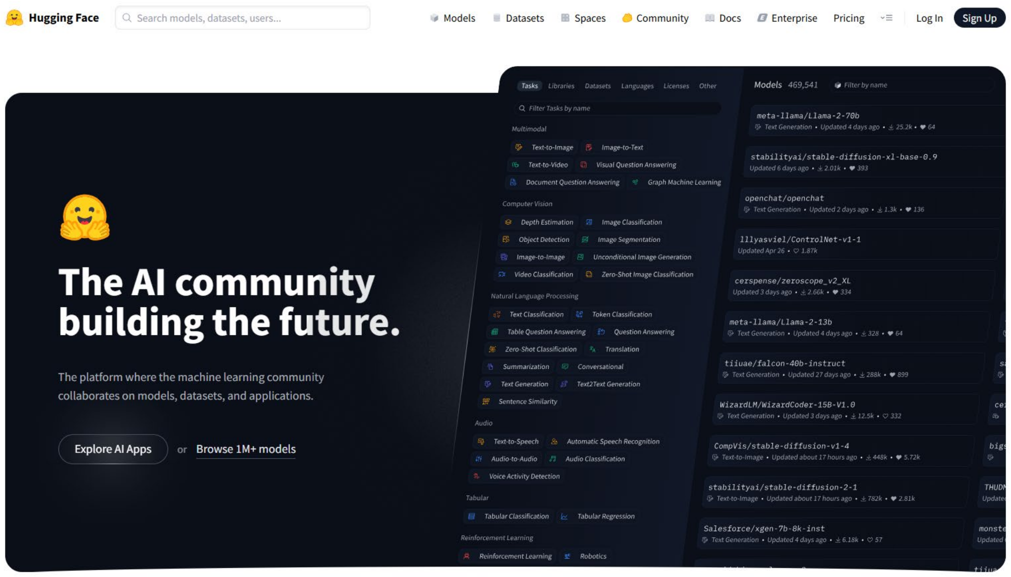
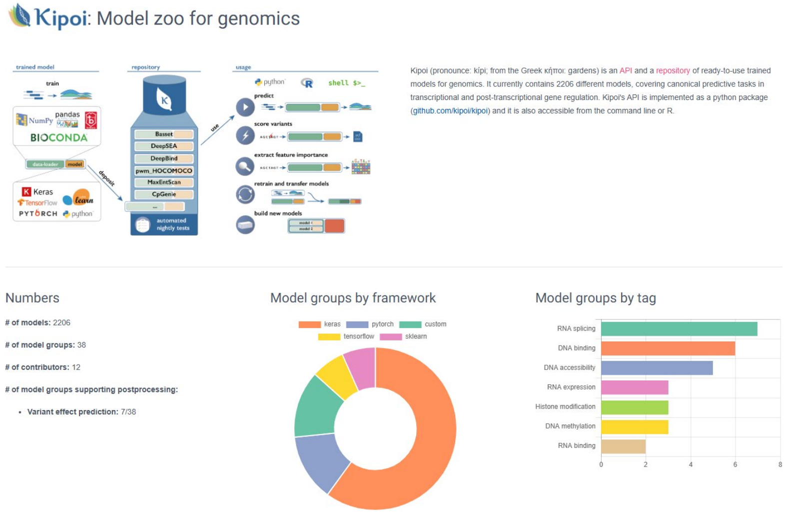
# Benefiting from available resources

- **Accelerated Development:** Leverage pre-trained models to jump-start projects, reducing prototyping and training time.
- **Resource Efficiency, Flexibility and Adaptability:** Utilize models that have already been trained on massive datasets, saving on computational costs and data collection efforts.
- **Robust Feature Extraction:** Benefit from learned representations that capture complex patterns, making fine-tuning more effective for specific tasks.
- **Standardized Benchmarks:** Use well-established models as baselines to ensure reproducibility and make comparative evaluations more straightforward.




https://huggingface.co/

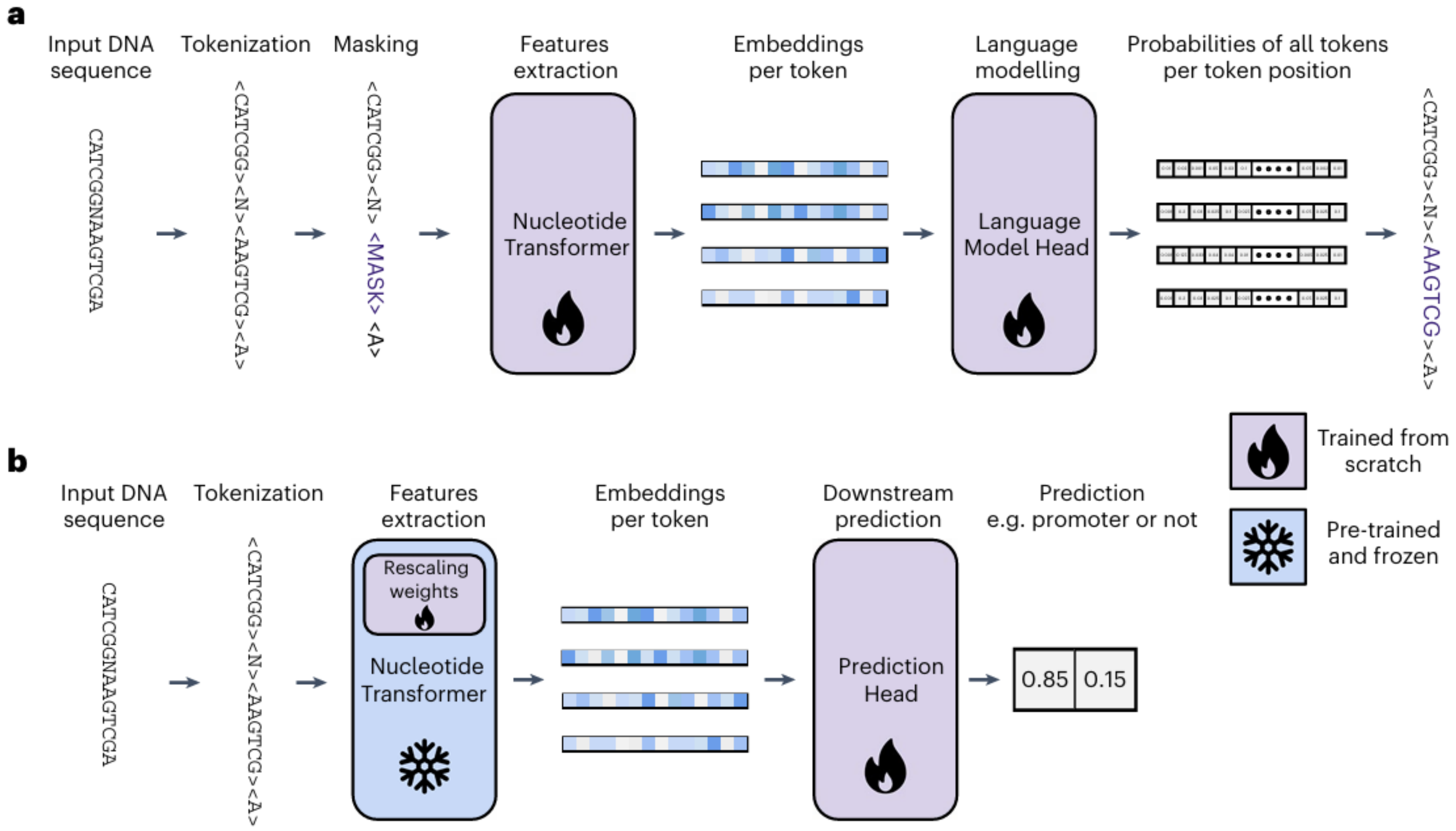



https://kipoi.org/

# Foundation models like DNABERT, HyenaDNA, NT or Enformer

**Foundation models are pre-trained on massive datasets, converting raw nucleotide sequences into rich, context-aware representations**

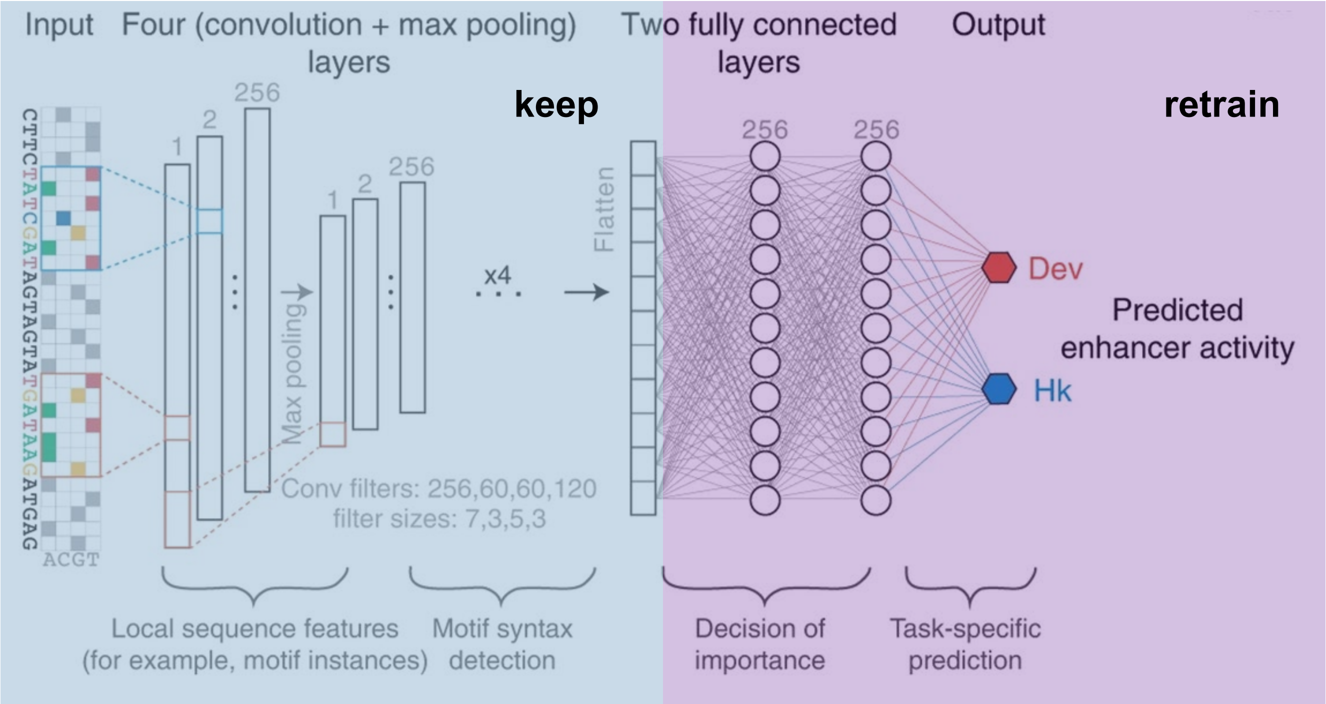
**Could refer specifically to language models (e.g. DNABERT, HyenaDNA, NT) but or could use all kinds of models.**

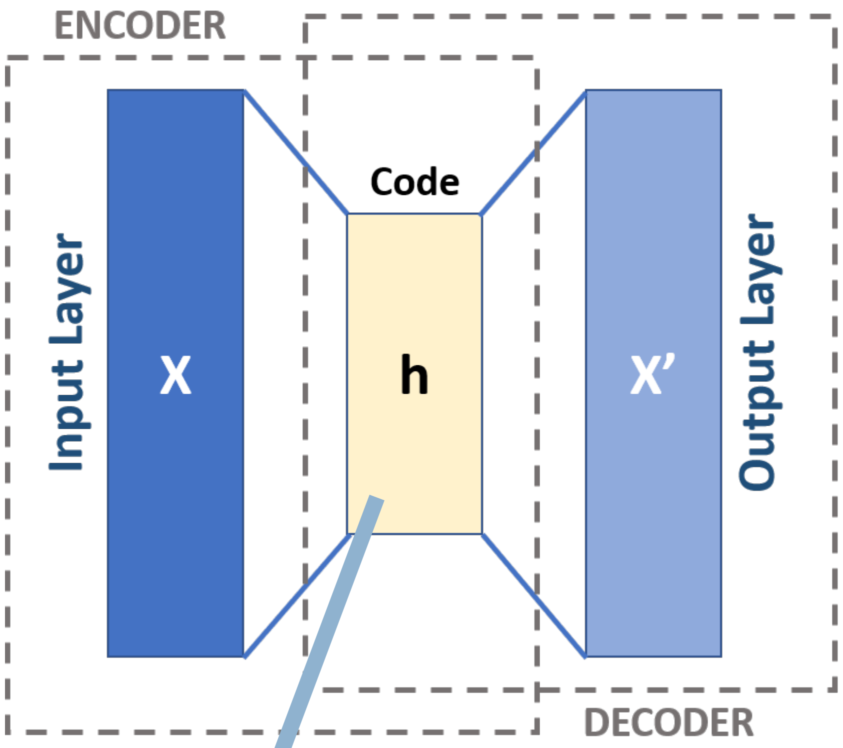



https://www.nature.com/articles/s41592-024-02523-z

# Using pre-trained layers, data transformation from autoencoders, ...

DeepSTARR




https://de.wikipedia.org/wiki/Autoencoder
#/media/Datei:Autoencoder_schema.png

**use latent variables as input to (classical) machine learning models**

# What is your experience or other insight?

# Feel free to reach out!


kircherlab.github.io

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# Hands-On sessions: Model interpretation

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# Hands-on: Interpreting models with in-silico mutagenesis

https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

**Go to:** 05_sequence_models/02_ism_and_tfmodisco.ipynb

**Klick:** Open in Colab

# Acknowledgements


- Pyaree M. Dash (BIH – Berlin, Germany)
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